TOUCHAUTH:A BIOMETRIC FIDO2 KEY

FIRST PHASE PROJECT REPORT

Submitted by

ASWINI SREEKUMAR **RIE22CC025**

NANDANA AJILAL **RIE22CC044**

to

APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree of

Bachelor of Technology (B.Tech)

in

Computer Science and Engineering (Cybersecurity)

Under the guidance of

Ms. SANDHYA K. R



Department of Computer Science and Engineering

Rajadhani Institute of Engineering and Technology Nagaroor,

Thiruvananthapuram

November, 2025

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DECLARATION

We undersigned hereby declare that the project report "**TOUCHAUTH:A BIOMETRIC FIDO2 KEY**", submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of **Ms. SANDHYA K.R**, **Assistant professor, Department of Computer Science & Engineering**. This submission represents my ideas in my own words and where ideas or words of others have been included. We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been previously formed the basis for the award of any degree, diploma or similar title of any other University

Place: NAGAROOR

Date: November, 2025

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the report entitled '**TOUCHAUTH: A BIOMETRIC FIDO2 KEY**' submitted by **ASWINI SREEKUMAR (RIE22CC025)** and **NANDANA AJILAL(RIE22CC044)** to the APJ Abdul Kalam Technological University for the partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by her under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ACKNOWLEDGEMENT

First and foremost, we express my heartfelt thanks and profound gratitude to the Management of Rajadhani Institute of Engineering and Technology and our Principal, **Dr. MADHUKUMAR S** for providing all the facilities to carry out this project work.

We am extremely happy to mention a great word of gratitude to **Dr. SANGEETHA SHIBU**, Head of department of computer science and engineering, for her valuable suggestions and encouragement to carry out this project work.

We would like to express my sincere gratitude to our guide **Ms. SANDHYA K. R**, Assistant Professor, Department of Computer Science and Engineering, for her relevant suggestions, guidance and inestimable leadership given for the completion of this project work. We thank our Project Coordinator **Ms. ASWATHY L C,** Assistant Professor, Department of Computer Science and Engineering, for her constant encouragement during the entire project work.

Finally, We take this opportunity to express my gratitude to the parents for their love, care and support and also to our friends and well-wishers who have been constant sources of support and inspiration for completing the project work.

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CO3	Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions of minimal complexity by using modern tools & advanced programming techniques (Cognitive Knowledge Level: Apply)
CO4	Prepare technical report and deliver presentation (Cognitive Knowledge Level: Apply)
CO5	Apply engineering and management principles to achieve the goal of the project (Cognitive Knowledge Level: Apply)

CO MAPPING TO POs

Mapping of course outcomes with program outcomes

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	3	3	3	2	2	2	2	2	2	2
CO2	3	3	3	3	3	3		2	2	2	2	2
CO3	3	3	3	3	3	2	2	2	2	2	2	2
CO4	3	3	3	3	3			2	2	2	2	2
CO5	3	3	3	3	3	2	2	2	2		2	2

ABSTRACT

The project titled “TouchAuth: A Biometric FIDO2 Key” presents an improvised and enhanced version of the authentication system developed during the first phase of the project. The objective of this system is to strengthen user authentication by combining biometric verification with modern passwordless authentication standards. By integrating f ingerprint recognition with the FIDO2/WebAuthn framework, the proposed solution aims to provide a more secure, user-friendly, and hardware-based authentication mechanism that reduces reliance on traditional passwords. The system architecture consists of an ESP8266 microcontroller, a fingerprint sensor, and an RP2040 Pico microcontroller functioning as a USB security key. The ESP8266 is responsible for biometric processing and fingerprint verification. When a user places a registered fingerprint on the sensor, the ESP8266 verifies the biometric template stored in the sensor module. Upon successful verification, the ESP8266 generates a virtual user presence signal through a GPIO interface and sends it to the RP2040 Pico. The RP2040 Pico then acts as a FIDO2 hardware authenticator, communicating with the host computer via USB. It performs secure cryptographic operations and completes the authentication process according to the FIDO2/WebAuthn standard, enabling passwordless login to supported systems and applications. This approach ensures that authentication requests are processed only after valid biometric verification. A key feature of the system is its modular architecture, which separates biometric processing from cryptographic operations. This separation enhances security by ensuring that sensitive information, such as fingerprint templates and private authentication keys, remains isolated within their respective components. As a result, the design minimizes the risk of data exposure while improving overall authentication reliability. The proposed system demonstrates a practical implementation of biometric-gated hardware authentication, offering improved protection against unauthorized access, phishing attacks, and password-related vulnerabilities.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	EXPANSIONS
AI	Artificial Intelligence
API	Application Programming Interface
CCID	Chip Card Interface Device
CTAP2	Client-to-Authenticator Protocol Version 2
ESP8266	Espressif Wi-Fi Microcontroller
FIDO	Fast Identity Online
FIDO2	Fast Identity Online 2 Authentication Standard
GPIO	General Purpose Input/Output
HAL	Hardware Abstraction Layer
HID	Human Interface Device
IDE	Integrated Development Environment
IoT	Internet of Things
LED	Light Emitting Diode
MCU	Microcontroller Unit
NFC	Near Field Communication
OTP	One-Time Password

RP2040	Raspberry Pi Microcontroller Chip
TRNG	True Random Number Generator
U2F	Universal Second Factor
UART	Universal Asynchronous Receiver-Transmitter
USB	Universal Serial Bus
WebAuthn	Web Authentication Standard

LIST OF LIBRARIES

1. ESP8266WiFi Library(ESP8266WiFi.h): Used to enable Wi-Fi functionality on the ESP8266 module. It allows the device to create a Wi-Fi Access Point (AP) so users can connect to the f ingerprint control web interface.
2. ESP8266WebServer Library(ESP8266WebServer.h): Used to host a web server on the ESP8266. It handles HTTP requests such as: /enroll /delete /clear /list /status. This enables the browser-based fingerprint management portal.
3. DNS Server Library(DNSServer.h):Used to create a captive portal.It redirects any domain request to the ESP8266 IP address so the fingerprint control webpage opens automatically.
4. Adafruit Fingerprint Library(Adafruit_Fingerprint.h):Used to communicate with the fingerprint sensor. It performs operations like: Capturing fingerprint images Converting image to template Matching fingerprints Storing fingerprint IDs Deleting fingerprints from memory.
5. SoftwareSerial Library(SoftwareSerial.h):Creates a software-based serial communication port on ESP8266 pins. This allows communication with the fingerprint sensor using GPIO pins instead of the hardware UART.
6. Standard C Libraries: These are basic C libraries used for common functions. stdio.h – Standard input/output functions , string.h – String manipulation functions , stdlib.h – Memory allocation and general utilities. These are used for functions like printf(), memcpy(), calloc(), etc.
7. Raspberry Pi Pico SDK Libraries:These libraries provide hardware support for the RP2040 (Raspberry Pi Pico). pico/stdlib.h – Standard Pico functions (timers, delays, GPIO utilities) pico/multicore.h – Multicore processing support pico/unique_id.h – Access unique chip ID pico/util/queue.h – Queue data structures for communication between tasks These libraries allow the program to control hardware and manage multitasking on the Pico.

8. TinyUSB Library: Used to implement USB functionality for the authentication token. tusb.h – TinyUSB stack for USB communication tinyusb.h – TinyUSB configuration for embedded systems This library enables the device to act as a USB HID / FIDO security key.

9. Raspberry Pi Pico Hardware Library: Used for direct hardware access. hardware/gpio.h – GPIO control functions This is used for reading the user presence button input.

10. Project-Specific Libraries / Headers: These are custom files used within your project. usb.h – USB interface definitions
usb_descriptors.h – USB device descriptor definitions
pico_keys.h – Main configuration for Pico Keys
apdu.h – Smart card communication commands
compat.h – Platform compatibility layer
user_presence.h – User presence verification functions.
These manage USB communication, authentication, and device configuration.

CHAPTER 1
INTRODUCTION

1.1 Overview

As cybersecurity threats continue to evolve in both scale and sophistication, the reliance on traditional password-based authentication has proven increasingly inadequate and vulnerable to exploitation. Passwords, once considered the cornerstone of digital identity protection, are now frequently targeted through phishing campaigns, replay attacks, brute-force attempts, and credential theft, all of which expose individuals and organizations to significant risks. Even with complex password policies and frequent updates, attackers have adapted their methods to bypass these defenses, leaving users frustrated and systems compromised. The limitations of passwords are not only a matter of security but also of convenience, as users often struggle to remember multiple complex credentials, leading to unsafe practices such as password reuse or reliance on weak combinations. This reality has created an urgent demand for stronger, more reliable, and user-friendly authentication mechanisms that can withstand modern threats while simplifying the login experience. Passwordless authentication has emerged as a transformative solution to these challenges, offering a paradigm shift in how digital identities are verified. By eliminating the need for memorized strings of characters, passwordless systems reduce the attack surface and remove the vulnerabilities associated with credential storage and transmission. Instead, these systems rely on advanced technologies such as biometric verification and hardware-based security keys, which provide cryptographic strength and ensure that authentication is tied directly to the unique physical presence of the user. This dual advantage of enhanced security and improved convenience makes passwordless authentication a compelling choice for modern digital ecosystems. The project presented here integrates several key technologies to deliver a secure and seamless authentication framework. Biometric authentication through fingerprint verification ensures that only legitimate users can initiate the login process, leveraging the uniqueness of fingerprints as a strong safeguard against impersonation. Hardware-based user presence control further strengthens this process by verifying that the user is physically present during authentication, thereby preventing remote exploitation or unauthorized access attempts. In addition, the system incorporates FIDO2 and WebAuthn cryptographic protocols, which are globally recognized standards for secure passwordless authentication and ensure interoperability across platforms and devices. USB security key functionality is also included, providing flexibility and resilience by enabling secure authentication in diverse in the

environments without requiring additional software installations.

Together, these components form a layered defense that addresses both security and usability, ensuring that authentication is executed only after successful fingerprint verification and that access is granted exclusively to authorized individuals.

The benefits of this integrated system are significant and extend beyond the immediate improvement in security. Enhanced protection is achieved by combining biometric verification with hardware presence, reducing the risk of credential theft and unauthorized logins. User convenience is improved by eliminating the need for passwords, streamlining the login process, and removing the frustration of forgotten or expired credentials. Compliance with modern standards is ensured through the adoption of FIDO2 and WebAuthn, aligning the system with industry best practices and regulatory requirements. The seamless experience offered by the system, which operates without requiring additional software installations or complex configurations, makes deployment straightforward and user-friendly. This combination of security and convenience is particularly important in today's digital environment, where users demand both protection and efficiency. By integrating fingerprint authentication, hardware-based presence control, cryptographic protocols, and USB key functionality, the project demonstrates how multiple technologies can work together to create a secure and user-centric authentication framework. The layered approach ensures that even if one component is challenged, the overall system remains resilient, providing confidence to both users and organizations. This resilience is critical in defending against modern cyber threats, which are increasingly sophisticated and persistent. The project not only addresses current vulnerabilities but also anticipates future challenges by adopting standards that are designed to evolve with the changing cybersecurity landscape. In doing so, it establishes a foundation for long-term digital security that balances the needs of users with the demands of organizations.

Beyond the immediate advantages of security and convenience, the broader implications of adopting passwordless authentication are profound, as it represents a decisive step toward reshaping the future of digital trust and identity management. By integrating biometric verification, hardware-based presence control, cryptographic standards, and USB key functionality into a unified framework, this project not only addresses current vulnerabilities but also anticipates the demands of tomorrow's interconnected world. As digital ecosystems expand to include cloud platforms, mobile applications, and Internet of

Things devices, the need for authentication systems that are both scalable and resilient becomes increasingly urgent. Passwordless solutions provide a foundation that can adapt to these emerging environments, ensuring that users remain protected regardless of the complexity of the systems they interact with. Moreover, the emphasis on interoperability through FIDO2 and WebAuthn standards ensures that the system can seamlessly integrate across diverse platforms, reducing fragmentation and enhancing user experience. This adaptability positions the project as more than just a technical innovation; it is a strategic response to the evolving landscape of cybersecurity, offering organizations a pathway to strengthen their defenses while empowering users with a frictionless and reliable means of access. In this way, the project not only delivers immediate value but also establishes a vision for sustainable digital security, where trust, usability, and resilience are harmonized to meet the challenges of an increasingly digital society.

In summary, the introduction of passwordless authentication supported by biometric verification and hardware security keys represents a major advancement in cybersecurity. This project demonstrates how the integration of fingerprint authentication, user presence control, cryptographic standards, and USB key functionality can deliver a secure, seamless, and user-centric login experience. By addressing the weaknesses of traditional password systems, the project lays the foundation for a future where digital identities are protected with stronger, more reliable methods. The emphasis on convenience ensures that users are not burdened by complex processes, while the focus on security guarantees that unauthorized access is effectively prevented. This balance between usability and protection is what makes the system truly innovative and practical. As organizations and individuals continue to navigate the challenges of the digital age, solutions like this project will play a crucial role in safeguarding sensitive information and maintaining trust in online interactions. The introduction sets the stage for a comprehensive exploration of how passwordless authentication can redefine cybersecurity practices, offering a vision of a digital environment where security and convenience coexist in harmony.

CHAPTER 2

LITERATURE SURVEY

2.1 Security Keys: Practical Cryptographic Second Factors for the Modern Web

Authors: M. Srinivas, A. Czeskis, D. Balfanz, B. T. Laurie, M. Marlinspike, and R. Sleevi

Overview: The paper "Security Keys: Practical Cryptographic Second Factors for the Modern Web" presents a comprehensive exploration of hardware-backed authentication as a solution to the vulnerabilities of passwords and OTPs. It introduces Security Keys that rely on challenge–response cryptography (ECDSA P-256, SHA-256) bound to the web origin, ensuring credentials cannot be phished or reused. The methodology is straightforward: during registration, the device generates a new key pair and provides the public key with attestation; during authentication, it signs a challenge with the private key, increments a counter to detect cloning, and enforces a Test of User Presence through a simple touch. The design emphasizes usability, deployability, privacy, and security, and evaluation includes performance benchmarks, comparative analysis against OTPs and smart cards, and large-scale deployment at Google with over 50,000 employees. Results show faster authentication, reduced support costs, and improved user satisfaction, with industry adoption by Dropbox and GitHub further validating its practicality.

Limitations: Despite these strengths, the paper acknowledges several limitations. Security Keys cannot guarantee transaction integrity since they lack a trusted display, leaving potential gaps if attackers manipulate browser content. Browser support was initially limited to Chrome, restricting immediate adoption. Attestation mechanisms balance privacy and revocability but sacrifice fine-grained device identification, while advanced schemes proved impractical for low-cost hardware. Hardware distribution costs remain a barrier, requiring multiple devices per employee and affordable pricing for consumers. Privacy concerns arise from global counters that may leak minor usage information, though per-key counters are harder to implement. Usability is improved but not flawless, as Security Keys still rely on passwords and require re-registration if lost. Finally, evaluation constraints limit generalizability, with phishing resistance measured indirectly and deployment data largely confined to Google's ecosystem. Overall, the paper demonstrates that Security Keys are a scalable, secure, and user-friendly second-factor solution.

2.2 U2F HID Implementation- Microprocessor to U2F Key

Authors: Ashikav Tareq

Overview: This work explores the design and implementation of a hardware-based Universal 2nd Factor (U2F) authentication key using a Teensy 3.2 microcontroller, demonstrating how low-cost embedded systems can replicate the security functions of commercial tokens. The methodology centers on embedding the U2F protocol stack directly into the microprocessor, enabling secure authentication through several layered mechanisms. AES-128 encryption is employed for key wrapping and application parameter protection, ensuring that sensitive data remains secure during communication. APDU framing is used to structure registration and authentication requests, while the HID protocol provides a communication channel between the microcontroller and browsers via WebAuthn. To prevent cloning, the device maintains a monotonic counter that increments with each authentication, allowing servers to detect anomalies. Trust is further reinforced through the use of an attestation certificate, which validates the authenticity of the device during registration. The prototype was tested successfully with modern browsers and Yubico's suite, confirming interoperability and functional correctness. This project illustrates that even resource-constrained microcontrollers can be adapted to deliver strong cryptographic authentication, bridging the gap between academic experimentation and practical deployment.

Limitations: While the prototype demonstrates feasibility, several limitations are acknowledged. First, the processing power and memory constraints of the Teensy 3.2 restrict scalability, making it unsuitable for more advanced cryptographic algorithms or larger deployments. The reliance on AES-128 encryption though secure today may not align with evolving standards that increasingly favor stronger or more flexible cryptographic primitives. Usability is another challenge: unlike polished commercial tokens, the prototype requires technical setup, firmware flashing, and lacks a user-friendly interface, which limits accessibility for non-technical users. Recovery mechanisms are underdeveloped; if the device is lost or corrupted, users must re-register credentials manually, which is cumbersome compared to enterprise-grade solutions. Compatibility is also limited, as the implementation was tested primarily with Chrome and Yubico tools, leaving cross-browser and cross-platform support uncertain. Finally, the evaluation scope is narrow, focusing on functional correctness rather than large-scale performance, usability, or resilience testing.

2.3 Building an Authentication Infrastructure - Designing a two factor authentication hardware token with form factor that encourages engagement

Authors: Zitao Zhang, Jacob Abbott, Sanchari Das, L. Jean Camp

Overview: This study explores the complex interplay between security, usability, and adoption in the design of modern authentication infrastructures. Recognizing that technical robustness alone does not guarantee widespread use, the authors adopt a mixed-methods approach that integrates both quantitative and qualitative research. A large-scale survey was conducted among university students who were provided with free YubiKeys, aiming to capture their perceptions of usability, convenience, and barriers to adoption. This survey revealed valuable insights into how younger, tech-savvy populations interact with hardware tokens, including their willingness to adopt them and the challenges they face in everyday use. Complementing the survey, the authors employed Research Through Design (RtD) to prototype a novel card-form hardware token embedded with a biometric sensor. This prototype was designed to investigate how physical form factors and biometric integration could influence user trust, acceptance, and perceived convenience. To evaluate the prototype, the researchers conducted usability workshops and a pilot Wizard-of-Oz study with seven participants, simulating real-world interactions with the device. These workshops provided qualitative insights into how users respond to authentication systems that combine hardware tokens with biometrics, highlighting both enthusiasm for stronger security and hesitation about carrying additional devices. By combining these methods, the paper emphasizes that authentication infrastructure design must extend beyond cryptographic strength to account for human behavior, institutional policies, and deployment environments. The findings suggest that while hardware tokens and biometrics can enhance security, adoption depends heavily on usability, accessibility, and integration into existing workflows.

Limitations: Despite its innovative approach, the paper acknowledges several limitations. The survey sample was restricted to a student population, which may not reflect the attitudes of enterprise employees or broader consumer groups. The Wizard-of-Oz testing involved only seven participants, limiting the depth and generalizability of usability insights. Furthermore, the prototype card-form token was not fully functional the biometric sensor was simulated rather than integrated meaning that results reflect user perceptions rather than actual performance under real-world conditions. The deployment environment was also constrained to academic settings, which differ significantly from enterprise-scale infrastructures where factors such as cost, distribution logistics, and recovery mechanisms are critical. Adoption barriers were evident: users expressed reluctance to carry extra hardware, concerns about cross-platform compatibility, and uncertainty about recovery processes in case of device loss. Additionally, the evaluation was primarily exploratory, focusing on perceptions and design possibilities rather than rigorous performance benchmarks or long-term field studies.In essence, while the paper demonstrates that combining surveys, prototyping, and usability testing can uncover both technical and human-centered challenges in authentication design, its limitations in scope, prototype maturity, and evaluation scale highlight the need for further research and real-world validation. Broader studies across diverse populations, fully functional prototypes, and enterprise-scale deployments will be essential to confirm the viability of such infrastructures and to refine them for widespread adoption.

2.4 User Authentication: Passive USB tokens, an alternative to Passwords?

Author: Kristoffer Rosvall Kristofer Niklasson

Overview: This study delves into the design and evaluation of authentication infrastructures that aim to strengthen digital identity systems while balancing security, usability, and scalability. The authors recognize that traditional password-based systems and even OTPs are increasingly inadequate in the face of phishing, credential theft, and sophisticated cyberattacks. To address these challenges, the paper explores hardware tokens, biometric authentication, and federated identity management as core components of a modern infrastructure. The methodology is multi-layered, combining literature review, system design, and prototyping. The literature review situates the work within the broader context of cybersecurity research, identifying gaps in existing authentication methods. The system design focuses on embedding authentication mechanisms into institutional workflows, ensuring that solutions are not only technically robust but also practically deployable. Prototyping efforts illustrate how biometric sensors and hardware tokens can be integrated into card-form factors, offering both enhanced security and user convenience. The paper also emphasizes the human factors of authentication, analyzing how user behavior, organizational policies, and regulatory frameworks shape adoption. By situating authentication within academic and enterprise contexts, the study underscores the importance of aligning cryptographic strength with usability and institutional trust.

Limitations: Despite its contributions, the paper acknowledges several limitations that temper its findings. The scope of evaluation is largely confined to academic and controlled environments, which restricts the generalizability of results to broader enterprise or consumer deployments. While conceptual models and prototypes are presented, fully functional implementations remain limited, leaving open questions about performance, scalability, and resilience under real-world conditions. Usability testing is relatively narrow, involving small participant groups, which constrains the depth of insights into adoption challenges across diverse demographics. Recovery mechanisms in the event of device loss or compromise are underdeveloped, raising concerns about continuity and user experience. Cross-platform compatibility is also uncertain, as the prototypes were tested in specific institutional contexts rather than across heterogeneous environments.

Another limitation lies in the institutional focus of the study. By centering on academic settings, the findings may not directly translate to global contexts where infrastructure, cost, and user literacy vary significantly. For example, while biometric integration and hardware tokens may be feasible in well-resourced institutions, they may pose challenges in regions with limited access to such technologies. Finally, the study does not fully address the long-term sustainability of authentication infrastructures, including lifecycle management of devices, regulatory compliance across jurisdictions, and evolving cryptographic standards. Additionally, the institutional focus narrows applicability: while biometric tokens may be feasible in well-resourced universities, they may be impractical in regions with limited access to such technologies. The study also does not fully address long-term sustainability, including lifecycle management of devices, regulatory compliance across jurisdictions, and evolving cryptographic standards. A further limitation lies in the lack of large-scale deployment data, meaning adoption barriers such as cost, distribution logistics, and user literacy remain speculative rather than empirically validated. Moreover, the research does not sufficiently explore user resistance and cultural factors issues such as privacy concerns with biometrics, reluctance to carry additional hardware, and varying levels of digital literacy could significantly affect adoption but are only lightly touched upon. Finally, the study's reliance on simulated prototypes (e.g., Wizard-of-Oz testing) means that user perceptions may not fully reflect real-world performance, leaving a gap between conceptual promise and practical implementation.

2.5 Modified USB Security Token for User Authentication

Authors: Wala'a M. AlOmari1 & Hesham Abusaimeh

Overview: This study presents a detailed exploration of how a modified USB-based security token can enhance user authentication by addressing the vulnerabilities inherent in password-only systems and traditional OTP mechanisms. The token is designed to resist phishing, credential theft, and replay attacks by employing cryptographic challenge–response protocols that tightly bind authentication to both the device and the service being accessed. The architecture integrates AES encryption for secure key wrapping, ensuring that sensitive data remains protected during communication. A monotonic counter is embedded to detect cloning attempts, providing an additional safeguard against duplication or tampering. Furthermore, attestation certificates are used to validate the authenticity of the device during registration, reinforcing trust between the token and the service provider. Communication between the token and host systems is achieved through USB HID protocols, which makes the solution compatible with existing browser infrastructures and WebAuthn frameworks. The paper emphasizes usability by simplifying the authentication process into a "plug-and-authenticate" interaction, reducing user burden compared to OTP entry or mobile-based verification. Testing demonstrates that the modified token can efficiently perform both registration and authentication, with performance benchmarks showing responsiveness suitable for real-world use. The authors highlight that this approach offers a low-cost alternative to commercial hardware tokens, making strong cryptographic authentication more accessible to institutions and individuals who may not have the resources for enterprise-grade solutions.

Limitations: Despite its promise, the paper identifies several limitations that must be addressed before such tokens can achieve widespread adoption. The prototype maturity is constrained by the hardware platform, with limited processing power and memory restricting scalability to more advanced cryptographic algorithms or multi-factor integrations. Usability remains a challenge, as the prototype requires technical setup, firmware flashing, and lacks the polished interface of commercial tokens, making it less accessible to non-technical users. Recovery mechanisms are underdeveloped if the token is lost, stolen, or corrupted, users must re-register credentials manually, which is cumbersome compared to enterprise-grade solutions that offer streamlined recovery workflows.

Compatibility is another concern. The prototype was tested primarily with specific browsers and Yubico tools, leaving cross-platform and cross-browser support uncertain. This raises questions about interoperability in diverse environments, particularly in enterprises that rely on heterogeneous systems. The evaluation scope is narrow, focusing on functional correctness rather than large-scale performance, usability, or resilience testing. Without stress-testing under adversarial conditions or enterprise workloads, it is difficult to assess how the token would behave in real-world deployments.

Cost and distribution logistics are also underexplored. While the token is presented as a low-cost solution, the paper does not fully address the challenges of manufacturing, distributing, and maintaining such devices at scale. Privacy concerns emerge from the use of global counters, which may inadvertently leak minor usage information, though implementing per-key counters is technically complex. Finally, the study does not fully explore **long-term sustainability**, including lifecycle management of devices, regulatory compliance across jurisdictions, and evolving cryptographic standards. These gaps highlight the need for further refinement and validation.

In essence, the paper demonstrates that a modified USB security token is a feasible, affordable, and secure authentication solution that can bridge the gap between academic experimentation and practical deployment. However, its limitations in scalability, usability, recovery, compatibility, and evaluation underscore the importance of broader usability studies, fully functional implementations, and enterprise-scale validation. Only through addressing these challenges can such prototypes evolve into robust, widely adopted authentication infrastructures capable of meeting the demands of modern digital identity systems.

CHAPTER 3

EXISTING SYSTEM

3.1 Existing system

In the current landscape of authentication technologies, most systems rely on traditional password- based verification, One-Time Passwords (OTPs), and proprietary hardware tokens such as YubiKey, YubiKey Bio (FIDO Edition), Google Titan Security Key, and SoloKeys. These devices follow FIDO2 and U2F (Universal 2nd Factor) standards that utilize asymmetric cryptography to protect user identities during login. In such systems, each authenticator generates a private-public key pair; the private key is securely stored in hardware, while the public key is registered with the service provider. During authentication, the private key signs a unique challenge from the server, thereby verifying the user's identity without transmitting or exposing sensitive credentials.

Although these systems represent a significant improvement over traditional passwords, they still rely heavily on PINs or touch confirmation rather than actual biometric verification. Most existing tokens serve as possession-based authenticators, proving ownership rather than personal identity. YubiKey and Titan keys, for example, provide high-grade security through encryption and challenge-response mechanisms but do not verify the individual user's identity beyond their physical access to the key. The systems are effective for phishing resistance and passwordless logins, but they do not ensure that the legitimate owner is the one performing the authentication. Furthermore, most available security keys are closed-source and proprietary, restricting customization, modification, or auditing. They are often expensive and depend on manufacturer-provided firmware that cannot be easily altered or integrated into open hardware development environments.

Another major drawback of existing systems is their reliance on external setup and recovery procedures, which are often complex for non-technical users. If a hardware token is lost or damaged, the user must go through multiple account recovery steps, sometimes involving cloud-based credential synchronization. This adds both dependency and vulnerability, especially for users seeking full control over their authentication methods. Additionally, these devices offer

limited adaptability , they are designed for specific use cases and lack the flexibility required for embedded applications, developer experimentation, or biometric enhancement. Hence, the current authentication ecosystem, while secure, lacks user-centric features, biometric verification, and customizable open-source alternatives. TouchAuth aims to address all these gaps by providing a compact, privacy-preserving, biometric authentication key built on open and verifiable hardware and software principles.

3.2 Advantages of Existing System

1. High Security:
   Existing FIDO2 and U2F devices like YubiKey and Titan Keys use public-key cryptography, ensuring that private keys never leave the hardware and cannot be stolen or duplicated.

2. Phishing and Replay Attack Resistance:
   Authentication is bound to a specific website's domain, preventing attackers from reusing authentication requests or credentials on fake sites.

3. Strong Two-Factor and Passwordless Authentication:
   These devices enable secure logins without relying on traditional passwords, combining hardware tokens with user presence verification for stronger access control.

4. Cross-Platform Compatibility:
   Supported across multiple operating systems (Windows, macOS, Linux, Android, ChromeOS) and browsers that comply with WebAuthn and FIDO2 standards.

5. Fast and Reliable Authentication:
   Hardware-based operations provide very low latency, ensuring quick response times compared to OTP or cloud-based systems.

6. Offline Functionality:
   Since FIDO2 tokens work over USB, NFC, or Bluetooth, they don't require an active internet connection for authentication, reducing network dependency.

7. Reduced Password Fatigue:
   Users are no longer required to remember or frequently reset complex passwords, improving overall user convenience and security hygiene.

8. Durability and Portability:
   Most hardware tokens are small, rugged, and easy to carry, making them suitable for professional and personal use across devices.

9. Enterprise-Grade Reliability:
   Widely adopted in industries requiring high-security logins, including government and corporate systems.

10. Regulatory Compliance:
    Hardware tokens fulfill modern cybersecurity standards and authentication policies required by global data protection regulations.

3.3 Disadvantage of Existing System

1. Proprietary Design:
   Most existing authentication tokens are closed-source, meaning users cannot view, modify, or audit the internal firmware or security processes.

2. Lack of Biometric Verification:
   Many hardware keys only verify user presence through a touch or PIN, without using biometric features like fingerprints or facial recognition.

3. Complex Setup and Recovery:
   The process of registering, configuring, and recovering tokens is often complicated and confusing, especially for non-technical users.

4. Risk of Token Loss:
   If the hardware token is lost or damaged, users may permanently lose access to accounts unless backup authentication methods are in place.

5. Vendor Dependency:
   Users rely heavily on manufacturers for firmware updates, security patches, and device recovery, creating dependency and trust issues.
6. Limited Customizability:
   The proprietary nature of most systems prevents developers from adding new features, integrating biometrics, or experimenting for research purposes.
7. Hardware Constraints:
   Tokens have limited storage and processing capabilities, which restrict the addition of new functionalities like continuous identity checks.
8. No Continuous Authentication:
   Existing systems only verify the user at login and cannot monitor ongoing user presence or behavior during a session.
9. Privacy Concerns:
   Some systems use cloud synchronization or third-party recovery services, increasing the risk of data exposure and reducing user control.
10. Compatibility Limitations:
    Certain hardware tokens are not fully compatible with older operating systems or legacy browsers, restricting their usability.
11. User Management Overhead:
    Managing multiple tokens for different accounts or recovery keys increases complexity and administrative burden for users.
12. High Cost:
    Commercial FIDO2 keys such as YubiKey and Titan are expensive, making them unaffordable for students, small businesses, and general users.
13. Limited Accessibility:
    Many advanced security keys are not easily available in all regions, and their high import cost or limited distribution restricts widespread adoption among general users.

CHAPTER 4

PROPOSED SYSTEM

4.1 Proposed system

Biometric authentication in this project is centered on fingerprint verification, which is one of the most widely adopted and trusted forms of identity confirmation. A fingerprint is unique to every individual, and unlike passwords, it cannot be guessed, stolen, or easily replicated. In this system, the fingerprint sensor captures the user's fingerprint and compares it against the stored template to confirm identity. Only when the fingerprint matches does the system allow the authentication process to proceed. This ensures that access is granted exclusively to the rightful user, eliminating risks associated with stolen or shared credentials. The use of biometric authentication also simplifies the login process, as users no longer need to remember complex passwords or carry additional information. It is a natural, intuitive, and secure way of proving identity, making it both user-friendly and highly reliable. By integrating fingerprint verification, the project ensures that authentication begins with a strong, unique, and personal safeguard that is difficult for attackers to bypass. Hardware-based user presence control adds another critical layer of protection by ensuring that the person attempting to log in is physically present during the authentication process. This mechanism prevents remote attacks where an intruder might try to gain access without being near the device. In simple terms, the system checks that the user is actually there, interacting with the hardware, before allowing authentication to continue. This is achieved through specialized hardware components that detect user interaction, such as pressing a button or touching a sensor, confirming that the login attempt is genuine. By requiring physical presence, the system blocks automated or remote exploitation attempts, making it much harder for attackers to compromise accounts. This feature is particularly important in environments where sensitive information is stored, as it ensures that even if credentials are somehow intercepted, they cannot be used without the user being physically involved in the process. The inclusion of hardware-based presence control strengthens the overall security posture of the project by combining human interaction with technological safeguards.

The project also incorporates FIDO2 and WebAuthn cryptographic protocols along with USB security key functionality, which together provide a modern, standardized, and highly secure method of authentication.

FIDO2 and WebAuthn are global standards designed to enable passwordless authentication across different platforms and devices. They work by using public key cryptography, where a private key is securely stored on the user's device and a corresponding public key is registered with the service. During login, the system verifies the user by checking the cryptographic challenge signed with the private key, ensuring that only the rightful device and user can authenticate. This process eliminates the need to transmit passwords, reducing the risk of interception or theft. The USB security key serves as a physical device that stores these cryptographic credentials and can be plugged into a computer or mobile device to complete the authentication process. It acts as a secure token, ensuring that even if someone knows the username, they cannot log in without the physical key. This combination of cryptographic standards and hardware keys provides strong protection against phishing, replay attacks, and credential theft, while also offering users a simple and convenient way to log in. Together, these components create a layered system where biometric verification, physical presence, cryptographic protocols, and hardware tokens work in harmony to deliver a secure, seamless, and user-centric authentication experience.

The project highlights the urgent need to move beyond traditional password systems by adopting passwordless authentication methods that combine biometric fingerprint verification, hardware-based user presence control, FIDO2 and WebAuthn cryptographic standards, and USB security key functionality. Together, these components create a secure and seamless login process that eliminates vulnerabilities such as phishing and credential theft while offering users a convenient and reliable experience. By balancing strong protection with ease of use, the system establishes a modern framework for digital identity management and sets the foundation for safer, more trustworthy interactions in an increasingly connected world.

The proposed system demonstrates how biometric authentication can be seamlessly integrated with hardware and software components to create a secure and user friendly access solution. By combining fingerprint recognition, USB communication, and user presence detection, the design ensures that authentication is both reliable and resistant to unauthorized use. The modular structure of the code allows each function to handle a specific task, from enrolling and verifying fingerprints to managing USB events and configurations. This separation of responsibilities makes the system easier to maintain and extend in the future, as developers can update or replace individual modules without disrupting the overall functionality. Beyond its technical robustness, the system emphasizes practical usability. Fingerprint enrollment and verification are designed to be intuitive, while USB descriptors and event handling ensure smooth communication with host devices. The inclusion of user presence detection adds an additional safeguard, requiring active participation from the user and reducing the risk of unattended or accidental access. These layered security measures reflect a thoughtful approach to balancing convenience with protection. The architecture also demonstrates scalability. New biometric modalities, additional USB interfaces, or enhanced event handling can be incorporated without major redesign, making the system adaptable to evolving requirements. Its efficiency in managing resources, such as queues and buffers, ensures that performance remains stable even under demanding conditions. This adaptability positions the system as a strong candidate for deployment in diverse environments, ranging from personal devices to enterprise level access control.

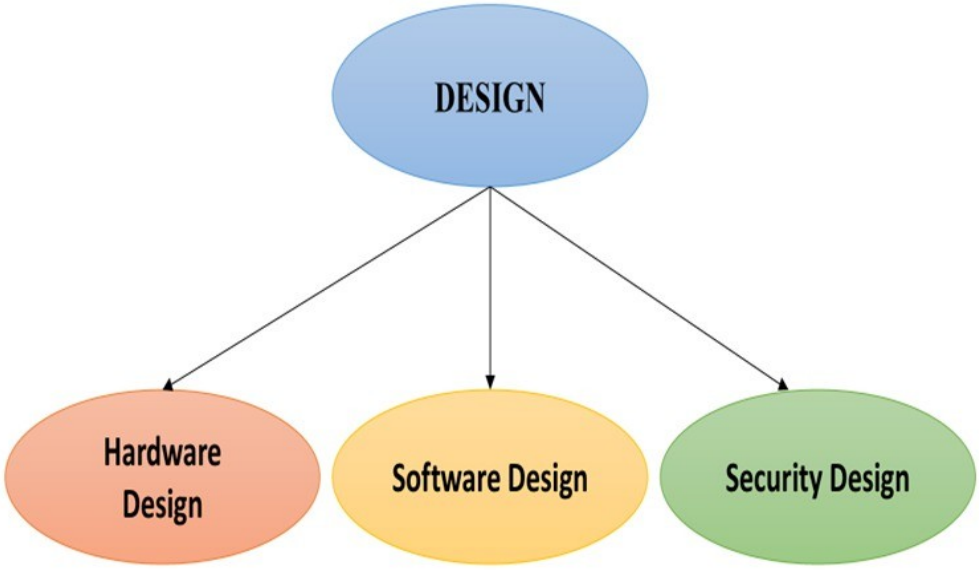
In conclusion, the proposed system is capable of delivering strong security while maintaining flexibility, scalability, and efficiency. It represents a practical and forward looking solution for environments where trust and controlled access are essential, offering a foundation that can grow with future technological advancements while continuing to safeguard user identity and data.

CHAPTER 5

DESIGN & IMPLEMENTATIONS

5.1 Design

5.1.1 Design Overview



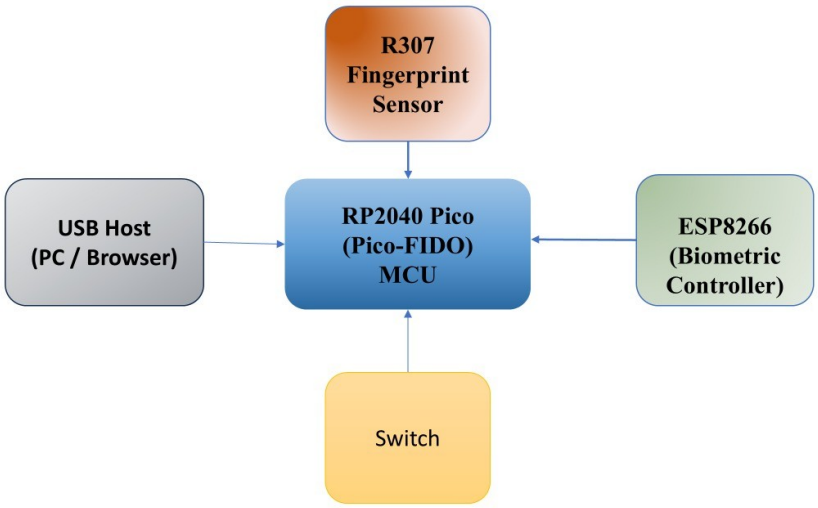
5.1.1.1 Design Structure Of Authentication System

This system is essentially a next-generation hardware security key that merges biometric fingerprint verification with FIDO2/WebAuthn standards to enable seamless, passwordless authentication across computers and online services. At its core, the design integrates a fingerprint sensor that ensures user presence and consent, coupled with a dedicated USB-based FIDO2 authenticator that functions like a modern hardware token. The architecture is carefully divided into three layers: hardware design, which focuses on the physical fingerprint module and secure USB interface; software design, which manages fingerprint enrollment, matching, and communication with authentication protocols; and security design, which enforces strong cryptographic protections, prevents spoofing, and guarantees that only verified users can trigger authentication. Together, these elements create a robust, user-friendly, and highly secure solution for passwordless access.

The system is designed as a hardware-based biometric security key that integrates fingerprint verification with FIDO2 and WebAuthn authentication to provide passwordless access to computers and web services, and its design can be understood through three interconnected layers: hardware, software, and security. At the hardware level, the system combines a fingerprint sensor with a dedicated USB FIDO2 authenticator, effectively emulating the functionality of a modern hardware security key. The fingerprint sensor captures the unique biometric data of the user, while the USB authenticator acts as the physical token that communicates with the computer or web service. This ensures that authentication only proceeds when the user is physically present and has successfully verified their fingerprint. On the software side, the system is programmed to manage the interaction between the fingerprint sensor and the FIDO2 authenticator, translating biometric input into secure authentication signals that comply with WebAuthn standards. The software ensures seamless communication between the hardware components and the applications or services being accessed, making the login process smooth and intuitive without requiring additional installations or complex configurations. Finally, the security design integrates cryptographic protocols that safeguard the authentication process against phishing, replay attacks, and credential theft. By using public key cryptography, the system ensures that sensitive data such as private keys never leave the device, and authentication challenges are securely signed and verified. This layered design, hardware for physical presence, software for seamless operation, and security protocols for cryptographic strength creates a robust framework that balances convenience with protection. The result is a passwordless authentication system that is simple to use, highly secure, and adaptable to modern digital environments, ensuring that users can access computers and web services with confidence and ease. This design not only strengthens the defense against modern cyber threats but also sets a foundation for scalable authentication solutions in the future. It demonstrates how hardware and software can be harmonized to deliver both usability and resilience. Ultimately, the system reflects a forward-looking approach to cybersecurity, where trust and convenience are equally prioritized.

5.1.2 Hardware Design:

The hardware architecture follows a biometric-gated authenticator model, where fingerprint verification provides the user-presence signal for FIDO2 authentication.



5.1.2.1 Block Diagram of Hardware Design

5.1.3 Hardware Interconnections:

Component	Interface	Purpose
Fingerprint Sensor → ESP8266	UART	Biometric verification
ESP8266 → Pico	GPIO	Virtual user-presence signal
Pico → PC	USB	FIDO2 authentication

5.1.3.1 Design Level Hardware Interconnections

1. ESP8266 to Fingerprint Sensor:

   The ESP8266 communicates directly with the fingerprint sensor through a UART serial link. When a user places their finger on the sensor, the module captures the fingerprint and performs template matching internally. Instead of sending raw biometric data, which could be insecure, the sensor only transmits a simple match or no-match result to the ESP8266. This ensures privacy while allowing the controller to confirm whether the fingerprint is valid.

2. ESP8266 to GPIO User Presence Signal:

   Once a fingerprint match is confirmed, the ESP8266 generates a GPIO signal to indicate user presence. This signal is critical because it acts as proof that the user is physically interacting with the system. The GPIO output is then passed to the RP2040 Pico, which uses it as a trigger to begin the authentication process.

3. ESP8266 and RP2040 Pico:

   The GPIO presence signal from the ESP8266 is directly connected to the RP2040 Pico, tightly coupling biometric verification with cryptographic authentication. This interconnection ensures that the Pico only performs FIDO2/U2F challenge–response operations after biometric verification has been completed. By linking these two modules, the system prevents unauthorized attempts to trigger authentication without a valid fingerprint.

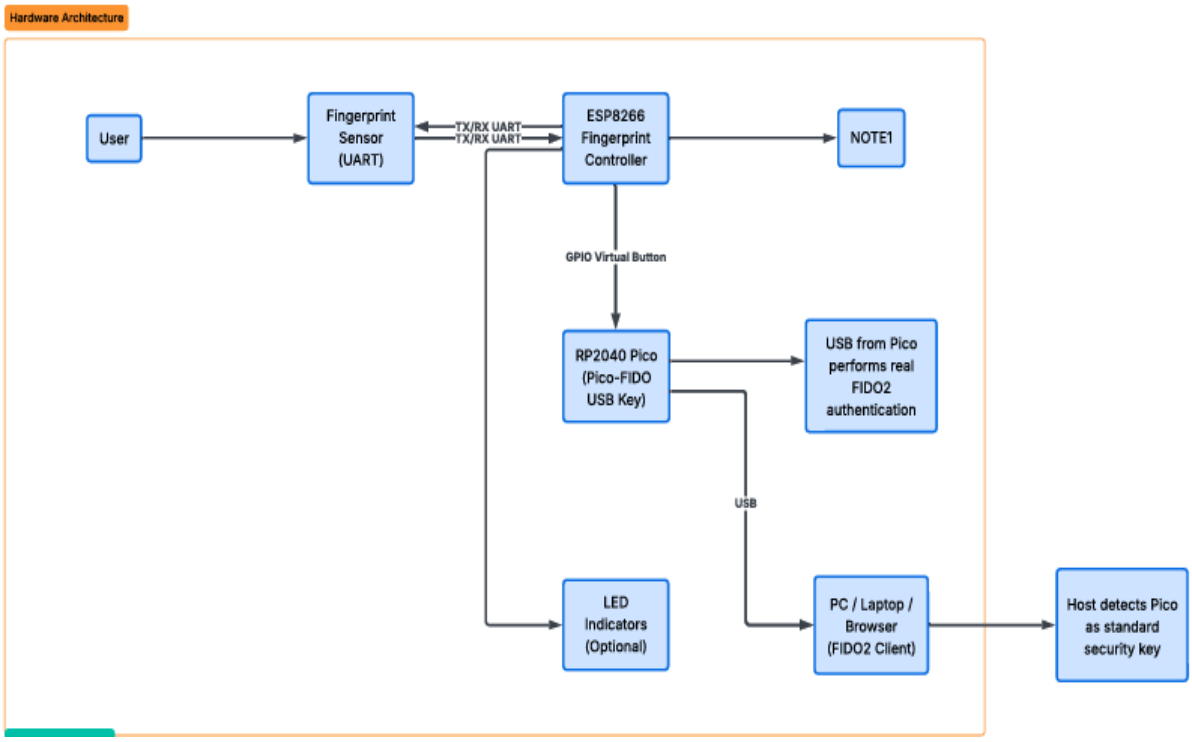
4. RP2040 Pico and USB Host (PC/Browser):

   The RP2040 Pico connects to the computer or browser via USB and enumerates as a standard FIDO2 security key. The host system communicates with the Pico using native WebAuthn/FIDO2 APIs, which are already supported by modern browsers and operating systems. This means no additional drivers or middleware are required, allowing seamless integration across platforms.

5. Authentication Flow Completion:

During login, the USB host sends a cryptographic challenge to the Pico, which responds by signing it with private keys securely stored in its memory. The host verifies this signed response against the registered public key, confirming the user's identity. Because the process is only triggered after the ESP8266 confirms fingerprint verification and user presence, the entire authentication flow is both secure and user-centric, combining biometric validation with cryptographic strength.

5.1.4 Hardware Architecture:

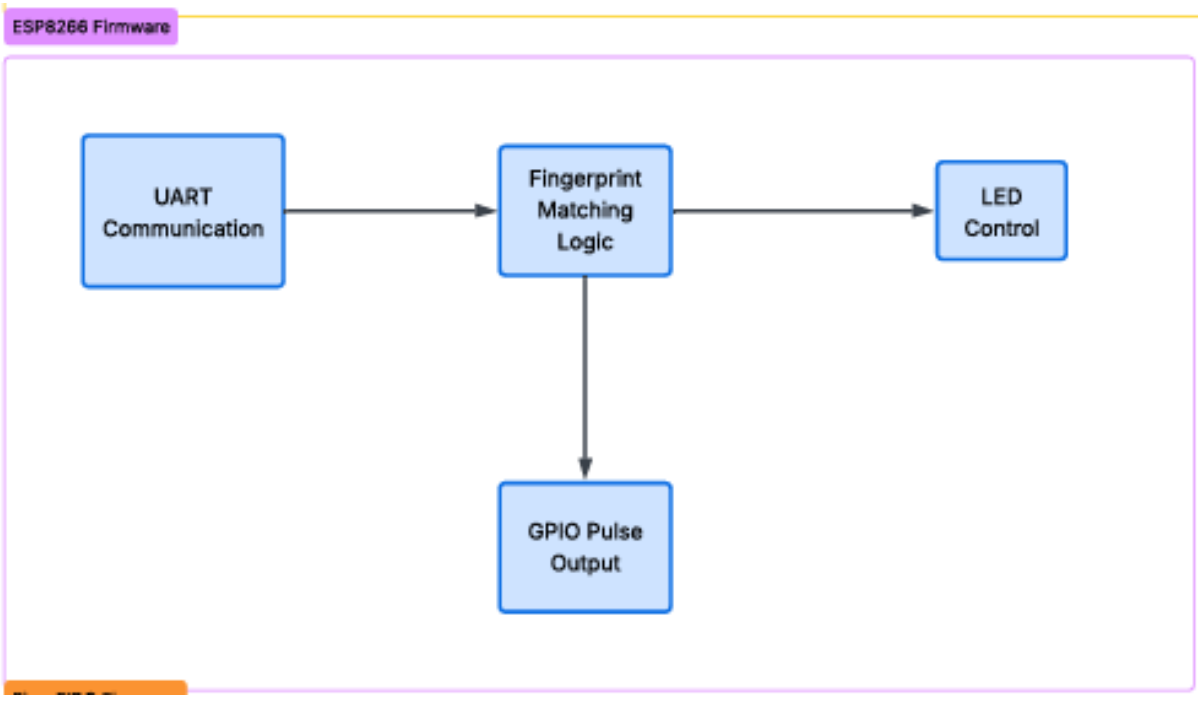


5.1.4.1 Logic Of Hardware Architecture

1. ESP8266 (Biometric Controller):

The ESP8266 serves as the brain of the biometric portion of the system. It is responsible for handling all communication with the fingerprint sensor through the UART interface, which is a reliable serial communication method. When a user places their finger on the sensor, the ESP8266 receives the match or no-match result and then generates a GPIO signal to confirm user presence. This signal acts as proof that the user is physically interacting with the device at that moment. In addition, the ESP8266 can control optional status LEDs, which provide immediate visual feedback such as showing when the system is ready, when a fingerprint has been accepted, or when authentication has failed. By managing both biometric input and user presence signaling, the ESP8266 ensures that authentication only proceeds when the rightful user is physically present and verified.

1.1 ESP8266 Firmware Logic



5.1.4.2 ESP8266 Firmware Block Diagram

The ESP8266 serves as the biometric-authorization controller, and its firmware is designed to manage all interactions with the fingerprint sensor while enforcing strict separation from cryptographic operations. When powered on, the ESP8266 initializes its UART interface, configures GPIO pins, and performs a handshake with the fingerprint sensor to confirm that the module is responsive and ready for biometric capture. Once initialization is complete, the ESP8266 enters an authentication-ready state, continuously monitoring the sensor for user interaction. When a finger is placed on the sensor, the ESP8266 receives a processed template-comparison result match or no-match over UART. The fingerprint sensor performs all biometric processing internally, ensuring that no raw fingerprint images or templates ever leave the sensor. If the result indicates a successful match, the ESP8266 generates a precisely timed GPIO pulse (typically 100–200 ms) that signals verified user presence to the Pico-FIDO authenticator. This pulse is the only form of communication between the ESP8266 and the Pico, ensuring that biometric verification is cleanly separated from cryptographic authentication.

The ESP8266 may also activate a success LED to provide immediate user feedback. If the fingerprint does not match, the ESP8266 rejects the attempt, optionally triggers a failure LED, and returns to its idle monitoring state. Throughout this process, the ESP8266's role is strictly limited to biometric authorization; it never handles, stores, or interacts with cryptographic keys or FIDO2 protocol data. This design ensures that the ESP8266 acts purely as a biometric gatekeeper, enforcing user presence while maintaining strong privacy boundaries and preventing any possibility of biometric data leakage into the authentication subsystem. The ESP8266 firmware is responsible for handling all biometric authorization tasks, beginning with the initialization of UART communication, GPIO configuration, and the fingerprint sensor itself. Once the system is ready, the ESP8266 enters an authentication-ready state where it continuously monitors the sensor for user interaction. When a fingerprint is detected, the sensor performs internal template matching and sends only a match or no-match result to the ESP8266. If the fingerprint is verified, the ESP8266 generates a controlled 100–200 ms GPIO pulse to signal user presence to the Pico and may activate a success LED for feedback.

If the fingerprint fails, the ESP8266 rejects the attempt, triggers a failure indicator if available, and returns to its idle monitoring state, ensuring it never participates in cryptographic operations.

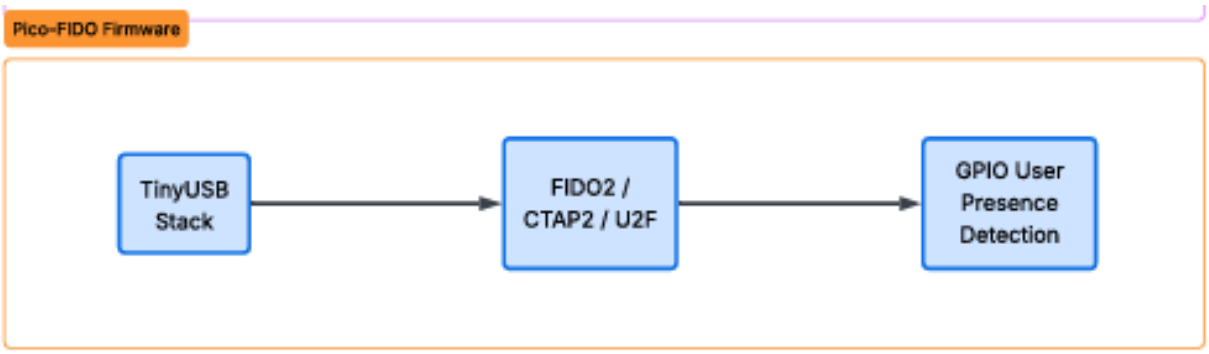
2. Match-on-Sensor Fingerprint Module:

The fingerprint sensor is an optical biometric module designed to capture and analyze the unique patterns of a user's fingerprint. Unlike simple image sensors, this module performs template matching internally, meaning it compares the captured fingerprint against stored templates within the sensor itself. Importantly, it does not expose raw biometric data to the rest of the system, which prevents sensitive information from being intercepted or misused. Instead, it sends a simple binary result match or no-match to the ESP8266. This design ensures that biometric data remains secure and private, while still allowing the system to reliably verify user identity. The sensor's ability to handle capture and matching internally reduces processing load on the controller and strengthens the overall security posture of the system.

3. RP2040 Pico (Pico-FIDO):

The RP2040 Pico acts as the dedicated USB FIDO2/U2F authenticator, essentially functioning as the hardware security key. When connected to a computer, it enumerates as a standard security key device that browsers and operating systems can recognize without additional drivers. Its primary role is to execute cryptographic challenge response operations, which are the foundation of passwordless authentication. The Pico securely stores private keys within its memory, ensuring they never leave the device. During authentication, the computer or browser sends a challenge, and the Pico responds by signing it with the private key, proving the user's identity without exposing sensitive data. This makes it resistant to phishing, replay attacks, and credential theft. By combining biometric verification from the ESP8266 with cryptographic operations on the Pico, the system ensures that authentication is both user-specific and cryptographically secure.

3.1 Pico-FIDO Firmware Logic



5.1.4.3 Pico-FIDO Firmware Block Diagram

The RP2040 Pico functions as the dedicated FIDO2/U2F authenticator, and its firmware is engineered to handle all cryptographic operations in a secure, isolated environment. Upon startup, the Pico initializes the TinyUSB device stack, allowing it to enumerate as a USB security key recognized by modern browsers and operating systems without additional drivers. The firmware implements the CTAP2 and U2F protocols, enabling the Pico to participate in WebAuthn registration and authentication flows. While idle, the Pico continuously monitors a designated GPIO pin for the user-presence pulse generated by the ESP8266.

This pulse acts as the Pico's confirmation that the user has been biometrically verified, satisfying the FIDO2 requirement for explicit user presence. Importantly, the Pico never receives, processes, or stores any fingerprint data; it only receives a binary presence signal. Once user presence is confirmed, the Pico proceeds to handle the authentication challenge issued by the host system. It retrieves the appropriate private key from its secure internal storage, signs the challenge using hardware-isolated cryptographic routines, and returns the signed response to the browser. The private keys never leave the Pico at any point, and all cryptographic operations occur entirely within the device, ensuring strong protection against key extraction, tampering, or interception.

This strict separation of responsibilities biometric verification handled by the ESP8266 and cryptographic authentication handled by the Pico creates a secure, modular architecture that aligns with FIDO2 security principles and guarantees that the authenticator remains trustworthy, privacy-preserving, and resistant to modern attack vectors. The Pico-FIDO firmware begins by initializing the TinyUSB device stack, allowing the device to enumerate as a standard FIDO2/U2F authenticator recognized by modern browsers. It implements the CTAP2 and U2F protocols, enabling secure communication with the host during registration and authentication. The Pico continuously monitors a dedicated GPIO pin for the user-presence pulse sent by the ESP8266, which serves as confirmation that biometric verification has already occurred. Once user presence is detected, the Pico processes the authentication challenge from the host and signs it using private keys stored securely within its internal memory. Throughout this process, the Pico never receives or handles fingerprint data, ensuring strict isolation between biometric processing and cryptographic authentication.

4. USB Host (PC/Browser):

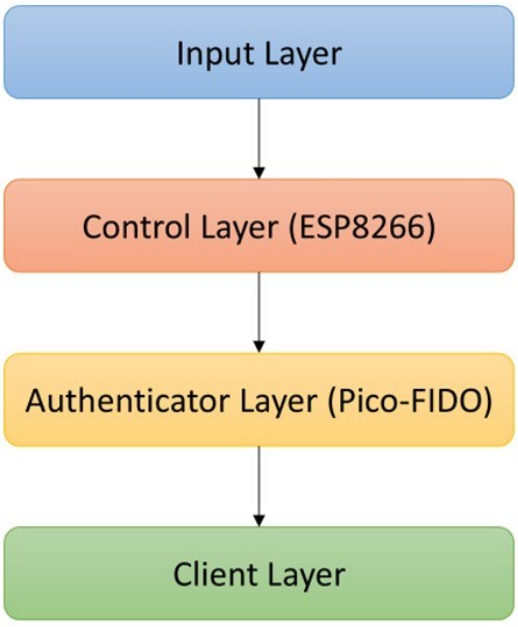
The USB host, typically a computer or web browser, plays the role of recognizing and interacting with the RP2040 Pico as a hardware security key. Modern browsers already support WebAuthn and FIDO2 APIs natively, which means the system works seamlessly without requiring any additional drivers or middleware. When a user attempts to log in, the browser communicates with the Pico to initiate the challenge response process. The host verifies the signed response against the stored public key, confirming that the authentication is valid. This integration ensures that the system can be used across different platforms and services without extra configuration, making it highly practical and user-friendly. The USB host essentially acts as the gateway between the user's hardware security key and the online service, ensuring smooth and secure communication during authentication.

5. Switch-based Mode Selection:

A physical switch is used to toggle between fingerprint registration mode and authentication mode.

- In Registration Mode, the switch connects the ESP8266 with the R307 fingerprint sensor, allowing the ESP8266 to handle the fingerprint enrollment process and store the fingerprint templates in the sensor memory.
- In Authentication Mode, control is switched to the RP2040 Pico running the Pico-FIDO firmware. In this mode, the Pico performs fingerprint verification and then executes the FIDO2/WebAuthn authentication with the host system through USB.This design separates the registration and authentication processes, ensuring secure biometric enrollment and controlled authentication.

5.1.5 Software Design:



5.1.5.1 Software Layer Architecture

1. Input Layer:

   The input layer is responsible for capturing the fingerprint and handling communication with the sensor. When a user places their finger on the optical fingerprint sensor, the module internally processes the fingerprint image and performs template matching against stored biometric data. Instead of exposing raw fingerprint images, which could compromise privacy, the sensor only sends a simple match or no-match result to the ESP8266 controller via UART communication. This design ensures that sensitive biometric information never leaves the sensor, while still allowing the system to reliably determine whether the fingerprint belongs to an authorized user. The input layer therefore acts as the first line of interaction, converting physical biometric input into a secure digital signal that can be processed further.

2. Control Layer (ESP8266):

   The ESP8266 functions as the biometric controller and decision-making unit. It receives the match/no-match result from the fingerprint sensor and determines whether authentication should proceed. If the fingerprint is verified, the ESP8266 generates a GPIO pulse lasting 100–200 milliseconds to indicate user presence. This pulse is critical because it signals to the next layer that the user is physically present and has passed biometric verification. The ESP8266 also manages authentication sequencing, ensuring that the process follows the correct order, and handles failure cases by rejecting attempts when the fingerprint does not match. In this way, the control layer acts as the gatekeeper, allowing authentication only after biometric success and preventing unauthorized or remote attempts to trigger the system.

3. Authentication Layer (Pico-FIDO):

   The RP2040 Pico, also referred to as Pico-FIDO, serves as the dedicated USB authenticator. It enumerates as a standard FIDO2/U2F security key when connected to a computer, meaning that operating systems and browsers recognize it as a hardware security device without requiring additional drivers. The Pico runs a USB stack (TinyUSB) and implements CTAP2/U2F protocols, which are the backbone of modern is passwordless authentication. When triggered by the GPIO pulse from the ESP8266, the Pico executes cryptographic challenge response operations. It securely stores private keys within its memory, ensuring they never leave the device, and uses them to sign authentication challenges sent by the host system. This guarantees that authentication is both cryptographically strong and resistant to phishing or replay attacks. The authenticator layer therefore transforms the biometric verification into a secure digital identity proof recognized by computers and web services.

4. Client Layer:

   The client layer consists of the computer or browser that interacts with the Pico-FIDO device. Modern browsers support WebAuthn APIs and operating systems include native FIDO2 security key support, so the client can seamlessly detect the Pico as a hardware security key. During login, the browser sends a challenge to the Pico, receives the signed response, and verifies it against the registered public key. This process completes the authentication flow, granting access only if the biometric verification and cryptographic challenge both succeed. Because the client layer relies on standardized APIs, the system works across multiple platforms and services without requiring additional middleware, making it highly practical and user-friendly.

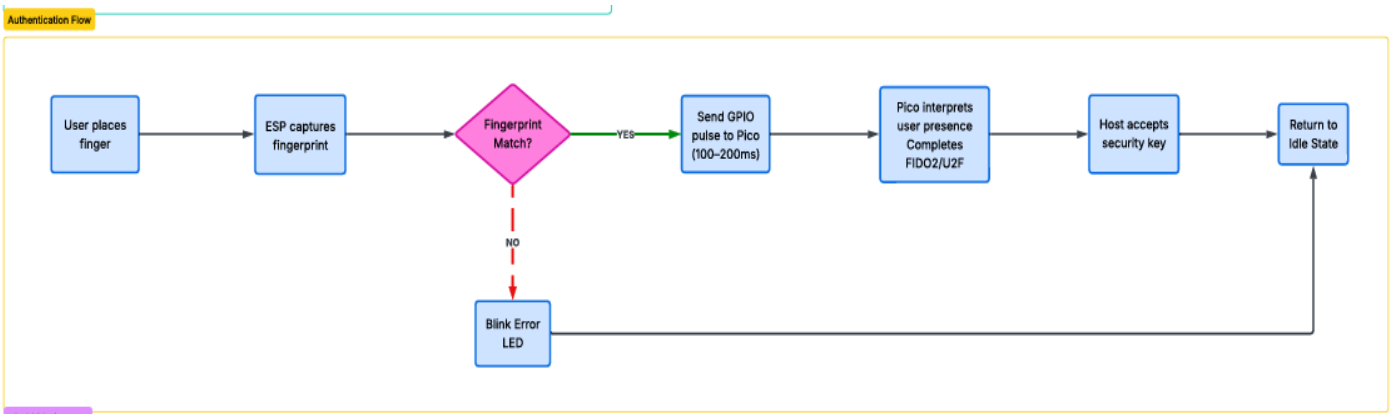
5. ESP8266 Firmware Design:

The firmware on the ESP8266 is the core of the biometric authentication pipeline, linking the fingerprint sensor to the Pico-FIDO authenticator. It handles UART communication to ensure accurate match/no-match results, and upon a successful fingerprint match, generates a 100–200 ms GPIO pulse that signals verified user presence. This pulse directly triggers Pico-FIDO's cryptographic operations, binding physical identity to digital security.

The firmware enforces strict failure logic: unmatched fingerprints immediately reject authentication, preventing any cryptographic sequence from proceeding without biometric success. Optional LED feedback can provide user cues, but the critical behavior is uncompromising authentication only occurs after biometric verification.

By embedding biometric checks into the workflow, the ESP8266 becomes the gatekeeper of trust, ensuring credentials are unlocked only when the rightful user is physically present. This integration of biometric verification with hardware-based cryptography strengthens security, blocks unauthorized access, and demonstrates how embedded firmware can serve as the cornerstone of resilient authentication infrastructures.

5.1.6 Security Design:



5.1.6.1 Fingerprint-Based Authentication Flow

This figure illustrates the complete flow of the fingerprint-based authentication process, showing how the system transitions from user interaction to final verification. The process begins when the user places a finger on the fingerprint sensor, which is managed by the ESP8266. The ESP8266 captures the fingerprint and performs template verification internally through the sensor module. At this decision point, if the fingerprint does not match, the ESP8266 immediately rejects the attempt and triggers an error LED to notify the user of failure. If the fingerprint matches, the ESP8266 generates a short GPIO pulse lasting 100–200 milliseconds, which is sent to the Pico-FIDO authenticator. The Pico interprets this pulse as verified user presence, a mandatory requirement in the FIDO2/U2F protocol, and then proceeds to complete the cryptographic authentication process. The host system, such as a browser or operating system, receives the signed response from the Pico and validates it against the registered public key. If the signature is valid, the host accepts the security key and grants access; if invalid, authentication is denied. Finally, regardless of success or failure, the system clears temporary states and returns to the idle state, ready for the next authentication cycle. This flow ensures that biometric verification, hardware signaling, and cryptographic proof are tightly integrated, providing both strong security and a clear, user-friendly process.

5.1.7 Authentication Model:

1. Someting you have – Physical Security Key:

   The first factor in the authentication model is the physical hardware security key, represented by the RP2040 Pico device. This module acts as a tangible token that must be present during every login attempt, ensuring that authentication cannot occur without the user physically possessing the key. By requiring this hardware component, the system prevents unauthorized remote access, since attackers cannot simply guess or steal credentials online. The physical key is recognized by computers and browsers as a standard FIDO2/U2F authenticator, meaning it integrates seamlessly with existing platforms. This "something you have" factor establishes a strong baseline of security by tying access to a physical device that is under the user's control.

2. Something you are – Fingerprint:

   The second factor is biometric verification, specifically fingerprint recognition. The fingerprint sensor captures and internally matches the user's fingerprint against stored templates, sending only a match or no-match result to the ESP8266 controller. This ensures that authentication is gated by the user's unique biometric identity, which cannot be guessed, shared, or easily replicated. By requiring a fingerprint match before the hardware key can proceed with cryptographic operations, the system guarantees that only the rightful user can activate the authenticator. This "something you are" factor adds a personal safeguard, making authentication both secure and intuitive, while also ensuring that user presence is genuine and not simulated remotely.

3. Cryptographic Proof – FIDO2 Private Keys:

   The third factor is cryptographic proof, achieved through private keys stored securely inside the authenticator's memory. These keys are generated during registration with the relying party and never leave the device, ensuring they cannot be intercepted or stolen. During login, the authenticator signs unique cryptographic challenges with the private key, and the server verifies the response using the corresponding public key. This process provides undeniable proof of identity and ensures that authentication is resistant to phishing and replay attacks.

By combining biometric verification with cryptographic challenge response operations, the system enforces a layered defense where both human presence and digital proof are required. This "cryptographic proof" factor completes the model, making authentication robust, trustworthy, and aligned with modern cybersecurity standards.

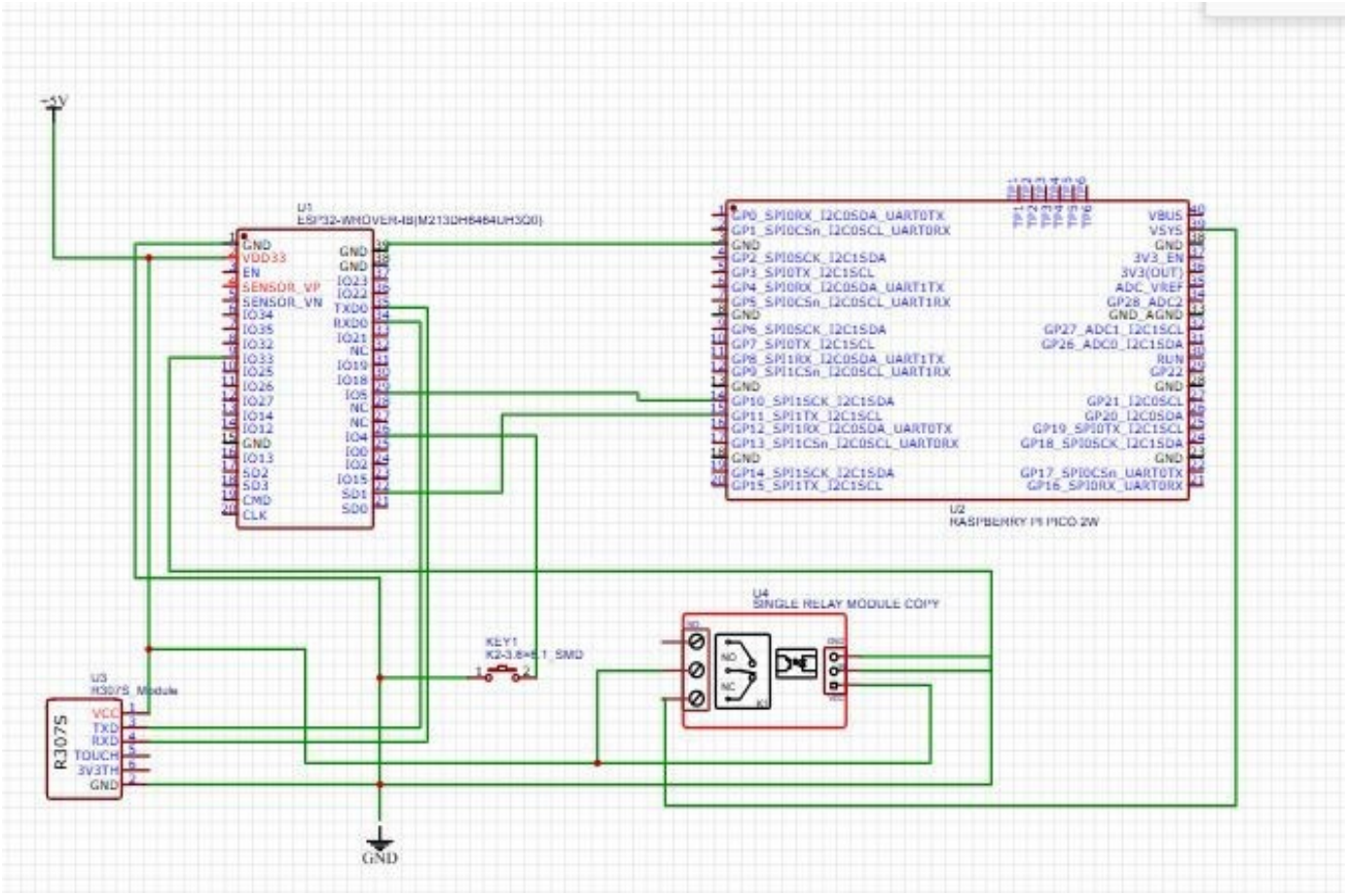
The outcome of the system design is a highly secure, modern, and user-friendly authentication solution that eliminates the weaknesses of traditional password-based systems. By integrating fingerprint verification with hardware-based access control and FIDO2 or WebAuthn cryptographic protocols, the design ensures that authentication is both passwordless and strongly protected against common cyber threats. The biometric-controlled user presence guarantees that only the rightful individual can activate the authenticator, while the physical security key enforces true hardware-based access control that cannot be bypassed remotely. At the same time, the system is built to work seamlessly with modern browsers and operating systems, requiring no additional drivers or middleware, which makes it practical and easy to deploy across different platforms. The reliance on cryptographic challenge response operations, combined with private keys stored securely inside the authenticator, provides strong defense against credential theft, phishing, and replay attacks. Overall, the design outcome is a layered authentication framework that balances convenience with robust protection, delivering a trustworthy solution for secure access to computers and web services in today's digital environment. In addition, the modular structure of the design makes it scalable, allowing future enhancements such as support for new biometric sensors or expanded cryptographic standards without disrupting the core functionality. This adaptability ensures that the system remains relevant and resilient as cybersecurity threats evolve and user expectations continue to rise. Furthermore, the design demonstrates how hardware isolation and biometric gating can work together to create a seamless yet uncompromising security experience. It also highlights the importance of privacy, as no biometric data ever leaves the device, reinforcing user trust in the system. Ultimately, the outcome reflects a forward-looking approach to authentication, setting a benchmark for secure, passwordless access in modern digital ecosystems.

5.2 Implementation:

The implementation of this project represents the practical realization of a secure, biometric-gated hardware authentication system built on the FIDO2 standard. At its core, the system integrates two distinct microcontroller platforms the ESP8266 and the RP2040 Pico each entrusted with specialized responsibilities that together form a layered security architecture. The ESP8266 firmware is designed to manage biometric input, initializing UART communication with the fingerprint sensor, capturing user interaction, and performing template verification internally through the sensor module. Upon successful fingerprint recognition, the ESP8266 generates a controlled GPIO pulse to signal verified user presence, while also providing immediate feedback through LED indicators. This signal is then interpreted by the Pico-FIDO authenticator, whose firmware implements the TinyUSB stack and the CTAP2/U2F protocols to handle cryptographic challenge–response operations. By isolating biometric processing from cryptographic execution, the implementation ensures that fingerprint data never leaves the sensor and private keys remain securely stored within the Pico, thereby enforcing strict privacy boundaries and hardware-anchored security. The workflow is designed to be cyclical and efficient: the system transitions from idle state to biometric capture, verification, cryptographic authentication, and finally resets to idle, ready for the next attempt. This modular approach not only simplifies debugging and testing but also makes the system scalable for future enhancements such as multi-user support, secure element integration, or wireless administrative management. Throughout the implementation, emphasis has been placed on seamless compatibility with modern browsers via WebAuthn, eliminating the need for additional drivers or middleware and ensuring a user-friendly experience. In practice, the implementation demonstrates how embedded hardware, firmware logic, and standardized security protocols can be harmonized to deliver a robust passwordless authentication solution that is both technically sound and operationally practical, laying the groundwork for deployment in enterprise, cloud, and high-security embedded environments.

5.2.1 Hardware Implementation:

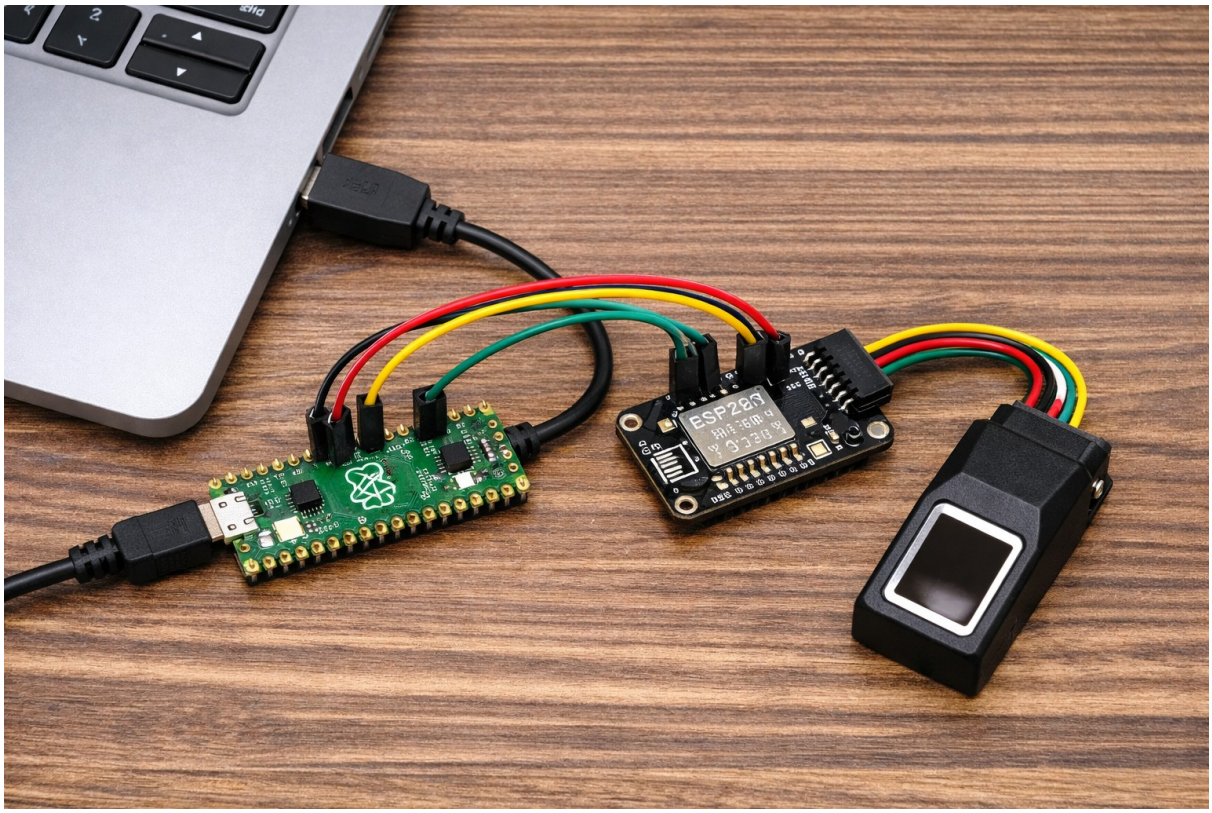
5.2.1.1 Schematic Diagram:



5.2.1.1.1 Pin Diagram

This circuit diagram shows a simple embedded setup where different components are connected to work together. The ESP8266 microcontroller is the main controller, linked to a touch sensor for user input and a relay module for switching external devices. Another supporting microcontroller handles communication through UART and I²C. The relay allows control of loads like appliances, while the sensor provides a trigger signal. Overall, it's a straightforward design combining microcontrollers, sensors, and actuators to enable basic automation.

5.2.1.2 Internal layout:



5.2.1.2.1 Hardware Setup for Biometric Authentication System

This figure shows the physical hardware setup used for the biometric authentication project. A laptop is connected via USB to a Raspberry Pi Pico board, which acts as the FIDO2 authenticator. The Pico is linked to an ESP8266 development board that manages the fingerprint sensor through jumper wires for power, ground, and data signals. The ESP8266 captures and verifies the fingerprint, while the Pico handles the cryptographic authentication process. Together, this arrangement demonstrates how the components are interconnected to deliver secure, passwordless access.

5.2.2 Software Implementation:

5.2.2.1 Core Fingerprint Processing Functions Code Snippets

1. Library Initialization and Hardware Setup

This snippet shows how the ESP8266 communicates with the fingerprint sensor using SoftwareSerial

```
#include <ESP8266WiFi.h>
#include <ESP8266WebServer.h>
#include <DNSServer.h>
#include <Adafruit_Fingerprint.h>
#include <SoftwareSerial.h>

#define AP_SSID "Fingerprint_Control"
#define AP_PASS "12345678"

SoftwareSerial mySerial(D5, D6);
Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);
```

5.2.2.1.1 ESP8266 Communication with the Fingerprint Sensor

This snippet initializes the required libraries for WiFi communication, web server hosting, DNS services, and fingerprint processing. The ESP8266 operates as a WiFi Access Point using the specified SSID and password. The fingerprint sensor communicates with the ESP8266 through a SoftwareSerial interface connected to GPIO pins D5 and D6.

2. ESP8266 Access Point and Web Server Setup

```
WiFi.mode(WIFI_AP);
WiFi.softAP(AP_SSID, AP_PASS);

webServer.on("/", []() {
  webServer.send(200, "text/html", portalPage());
});

webServer.on("/enroll", handleEnroll);
webServer.on("/delete", handleDelete);
webServer.on("/clear", handleClear);
webServer.on("/list", handleList);
webServer.begin();
```

5.2.2.1.2 Fingerprint Control Web Server Code

This section configures the ESP8266 to operate in Access Point mode and hosts a web interface for controlling the fingerprint system. The web server defines different routes to handle operations such as fingerprint enrollment, deletion, clearing the database, and listing stored fingerprints.

3. Fingerprint Sensor Initialization

This snippet verifies that the fingerprint sensor is connected and ready.

```
mySerial.begin(57600);
finger.begin(57600);

if (!finger.verifyPassword()) {
    while (1);   // stop if sensor not found
}
```

5.2.2.1.3 Fingerprint Sensor Initialization Code

The ESP8266 establishes communication with the fingerprint module at 57600 baud rate. The function verifyPassword() checks whether the fingerprint sensor is correctly connected and authenticated. If the sensor is not detected, the system stops execution to prevent further errors.

4. Fingerprint Verification (Authentication)

This is the main fingerprint matching logic.

```
void verifyFinger() {
    if (finger.getImage() != FINGERPRINT_OK) return;
    if (finger.image2Tz() != FINGERPRINT_OK) return;

    if (finger.fingerSearch() == FINGERPRINT_OK) {
        lastStatus = "Fingerprint matched → ID " + String(finger.fingerID);

        if (digitalRead(button_pin) == LOW) {
            digitalWrite(relay_pin, LOW);
            delay(60000);
            digitalWrite(relay_pin, HIGH);
        }
    }
    else {
        lastStatus = "No fingerprint match";
    }
}
```

5.2.2.1.4 Fingerprint Verification and Relay Control Code

This function continuously scans the fingerprint sensor. When a finger is placed, the image is captured and converted into a digital template. The template is compared with stored fingerprints in the sensor database. If a match is found, the corresponding ID is retrieved and the relay is activated, allowing the system to proceed with authentication.

5. Fingerprint Enrollment Function

This snippet registers a new fingerprint.

```
uint8_t getFingerprintEnroll() {
    int p = -1;

    while (p != FINGERPRINT_OK) {
        p = finger.getImage();
    }

    finger.image2Tz(1);
    delay(2000);

    while (finger.getImage() != FINGERPRINT_NOFINGER);

    while (finger.getImage() != FINGERPRINT_OK);

    finger.image2Tz(2);

    if (finger.createModel() != FINGERPRINT_OK)
        return false;

    return finger.storeModel(id) == FINGERPRINT_OK;
}
```

5.2.2.1.5 Fingerprint Enrollment Function Code

The enrollment process captures the fingerprint image twice to ensure accuracy. Both images are converted into templates and combined to create a fingerprint model. This model is then stored in the fingerprint sensor's internal database with a unique ID for future authentication.

6. Deleting Fingerprints

This snippet removes a fingerprint from the database.

```
void handleDelete() {
    uint8_t d = webServer.arg("id").toInt();
    finger.deleteModel(d);
}
```

5.2.2.1.6 Fingerprint Deletion Handler Code

This function deletes a stored fingerprint template from the sensor memory using its corresponding ID. This allows administrators to remove users from the authentication system when required.

Raspberry Pi Pico Firmware Implementation Code Snippets

1. User Presence Detection Using GPIO Button

```
#include "user_presence.h"
#include "hardware/gpio.h"
#include "pico/stdlib.h"

#define USER_BUTTON_GPIO 15   // GPIO pin for button

void user_presence_init(void) {
    gpio_init(USER_BUTTON_GPIO);
    gpio_set_dir(USER_BUTTON_GPIO, GPIO_IN);
    gpio_pull_up(USER_BUTTON_GPIO);   // active LOW
}

bool user_presence_wait(uint32_t timeout_ms) {
    absolute_time_t start = get_absolute_time();

    while (absolute_time_diff_us(start, get_absolute_time()) < timeout_ms * 1000) {
        if (gpio_get(USER_BUTTON_GPIO) == 0) {
            sleep_ms(40); // debounce
            return true;
        }
        tight_loop_contents();
    }
    return false;
}
```

5.2.2.1.7 User Presence Detection Code

This code implements user presence verification using a push button connected to a GPIO pin of the microcontroller. The function user_presence_init() initializes the button pin as an input with an internal pull-up resistor, making the default state HIGH and detecting a button press when it becomes LOW. The function user_presence_wait() monitors the button for a specified timeout period and includes a 40 ms debounce delay to prevent false triggers. If the button is pressed, the function returns true, confirming user presence; otherwise, it returns false. This mechanism ensures that the authentication process proceeds only when a physical user interaction is detected, thereby improving system security.

2. USB Device Descriptor (Device Identity Explanation)

```
tusb_desc_device_t desc_device = {
    .bLength            = sizeof(tusb_desc_device_t),
    .bDescriptorType    = TUSB_DESC_DEVICE,
    .bcdUSB             = (USB_BCD),

    .bDeviceClass       = 0x00,
    .bDeviceSubClass    = 0,
    .bDeviceProtocol    = 0,
    .bMaxPacketSize0    = CFG_TUD_ENDPOINT0_SIZE,

    .idVendor           = (USB_VID),
    .idProduct          = (USB_PID),
    .bcdDevice          = PICO_KEYS_SDK_VERSION,

    .iManufacturer      = 1,
    .iProduct           = 2,
    .iSerialNumber      = 3,

    .bNumConfigurations = 1
};
```

5.2.2.1.8 USB Device Descriptor Definition Code

This structure defines the USB device descriptor, which identifies the hardware authentication token to the host computer. It includes vendor ID, product ID, manufacturer information, and USB version used by the device.)

3. USB Initialization Function

```
void usb_init()
{
    mutex_init(&mutex);

    queue_init(&card_to_usb_q, sizeof(uint32_t), 64);
    queue_init(&usb_to_card_q, sizeof(uint32_t), 64);

    uint8_t enabled_usb_itf = PHY_USB_ITF_CCID | PHY_USB_ITF_WCID |
                              PHY_USB_ITF_HID | PHY_USB_ITF_KB;
}
```

5.2.2.1.9 USB Initialization Function Code

The usb_init() function initializes the USB subsystem. It sets up message queues used for communication between the USB interface and the authentication module, enabling efficient event handling during the authentication process.

4. USB Task Handler

```
void usb_task() {
#ifdef USB_ITF_HID
    hid_task();
#endif

#ifdef USB_ITF_CCID
    ccid_task();
#endif
}
```

5.2.2.1.10 USB Task Execution Code

The usb_task() function continuously processes USB communication tasks. It handles FIDO HID communication and smart-card interface communication, allowing the authentication token to interact with the host system.

5. USB Event Communication

```
void usb_send_event(uint32_t flag) {
    queue_add_blocking(&usb_to_card_q, &flag);

    if (flag == EV_CMD_AVAILABLE) {
        timeout_start();
    }

    uint32_t m;
    queue_remove_blocking(&card_to_usb_q , &m);
}
```

5.2.2.1.11 USB Event Handling Code

This function sends event signals between the USB thread and the authentication processing thread. It enables synchronized communication between the USB interface and internal authentication modules.

6. USB Event Definitions

```
#define EV_CARD_CHANGE        1
#define EV_TX_FINISHED        2
#define EV_EXEC_ACK_REQUIRED  4
#define EV_EXEC_FINISHED      8
#define EV_RX_DATA_READY     16
#define EV_PRESS_BUTTON      32
```

5.2.2.1.12 Event Flag Definitions Code

These macros define event flags used by the firmware to manage USB communication states such as receiving data, transmitting responses, and detecting user interaction events.

7. USB Data Buffer Structure

```
typedef struct {
    uint8_t buffer[USB_BUFFER_SIZE];
    uint16_t r_ptr;
    uint16_t w_ptr;
} usb_buffer_t;
```

5.2.2.1.13 USB Buffer Structure Definition Code

This structure represents the USB communication buffer used to temporarily store data packets transmitted between the host system and the authentication token. The read and write pointers help manage buffer operations efficiently.

8. USB Stack Configuration

```
#define CFG_TUD_ENABLED       1
#define CFG_TUD_MAX_SPEED     BOARD_TUD_MAX_SPEED
#define CFG_TUD_ENDPOINT0_SIZE    64
```

5.2.2.1.14 USB Configuration Settings Code

These configuration parameters define the TinyUSB stack settings used by the device, including endpoint size, maximum communication speed, and enabling the USB device stack.

9. CCID Smart Card Descriptor Structure

```
struct ccid_class_descriptor {
    uint8_t  bLength;
    uint8_t  bDescriptorType;
    uint16_t bcdCCID;
    uint8_t  bMaxSlotIndex;
    uint8_t  bVoltageSupport;
};
```

5.2.2.1.15 CCID Class Descriptor Structure Code

This structure defines the CCID smart-card class descriptor, which allows the authentication token to emulate smart-card functionality for secure communication with the host system

CHAPTER 6

RESULT

6.1 Result

The results highlight the successful working of the TouchAuth system, where fingerprint verification through the ESP8266 and R307 sensor is combined with secure authentication handled by the RP2040 Pico. Testing confirmed that biometric input is captured and validated locally, then translated into a user-presence signal for FIDO2 authentication. The login process, including fingerprint registration and two-factor authentication, worked smoothly, showing that the separation of biometric and cryptographic operations ensures both privacy and strong security. Overall, the outcomes demonstrate that the project delivers fast, user-friendly, and reliable passwordless authentication.



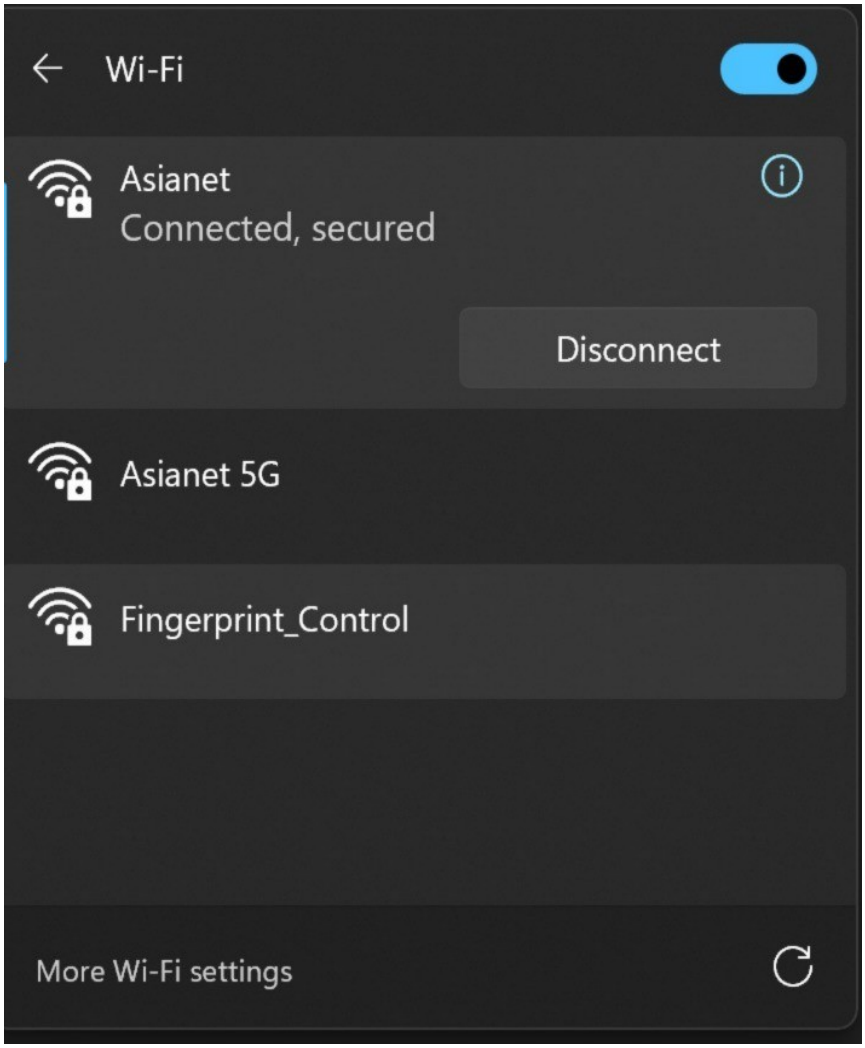
6.1.1 TouchAuth Device



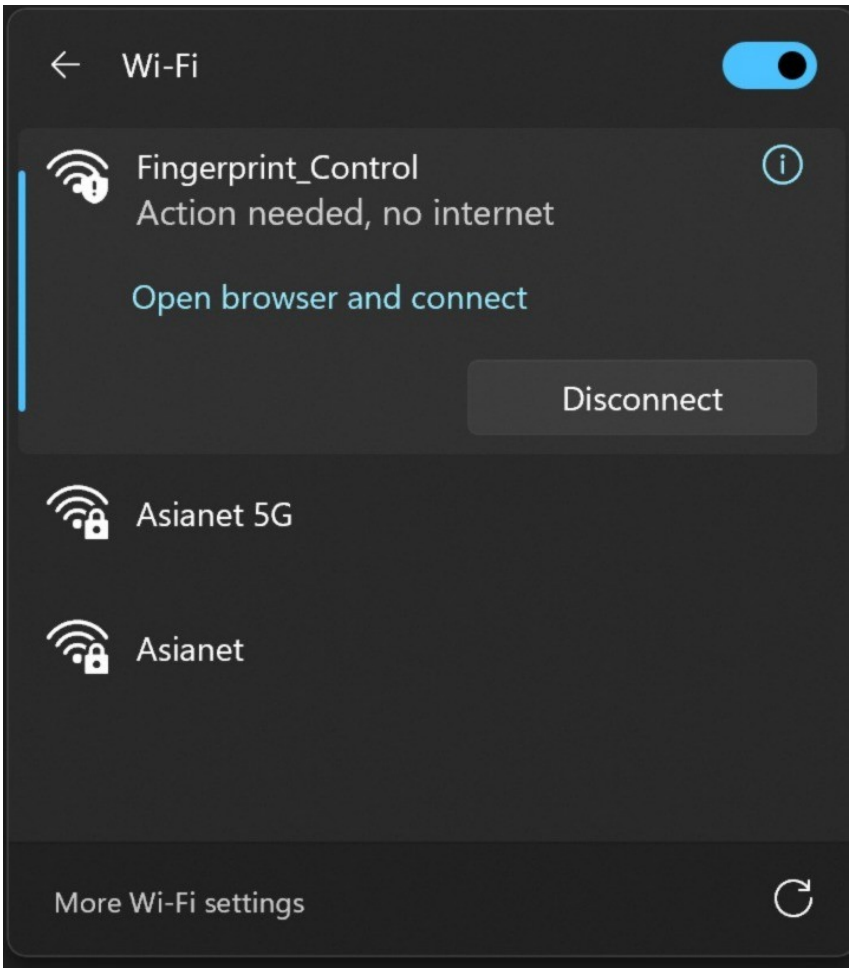
6.1.2 TouchAuth Device Plugged-In

6.2 Fingerprint Registration

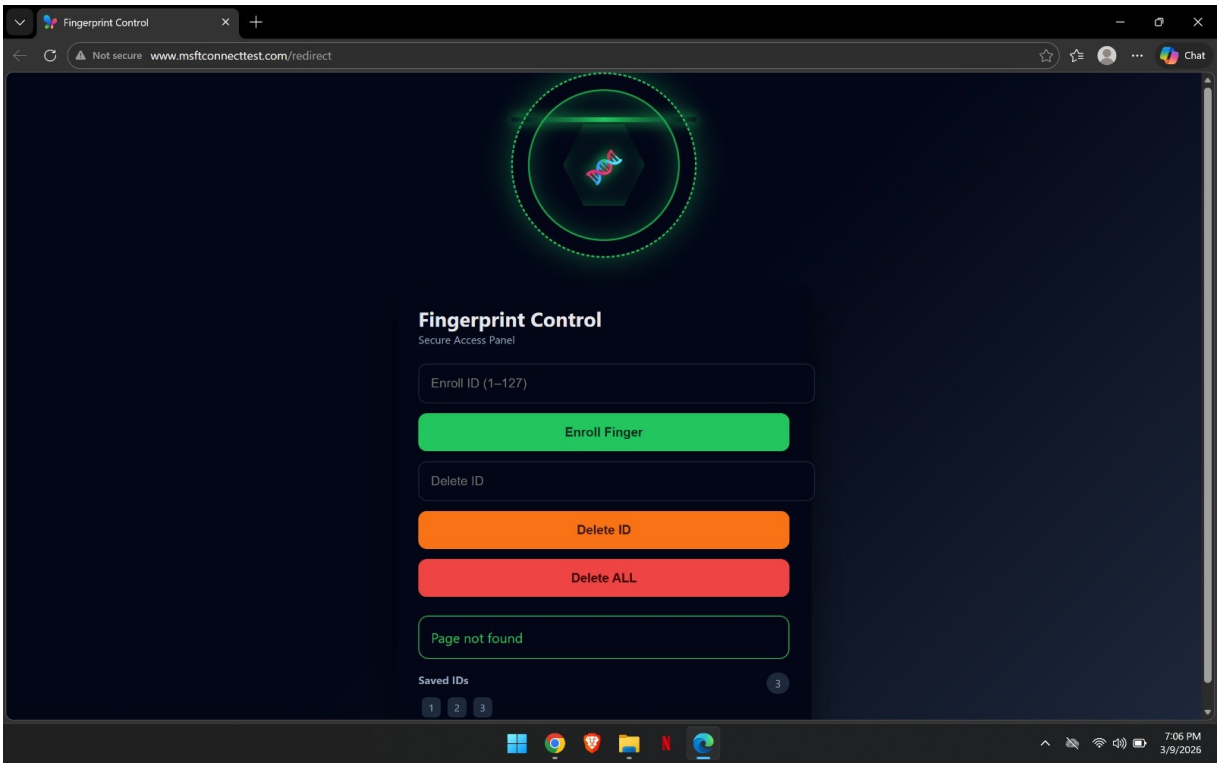
These figures show the Wi-Fi connectivity process during fingerprint registration. In the first image, the ESP8266 creates a dedicated access point named *Fingerprint_Control*, which appears alongside other available networks. In the second image, once connected to this access point, the system prompts the user to open a browser for enrollment, indicating that fingerprint registration and management are handled locally through a secure web interface. This ensures that biometric data is registered safely before being used for authentication.



6.2.1 Device Detected By WIFI



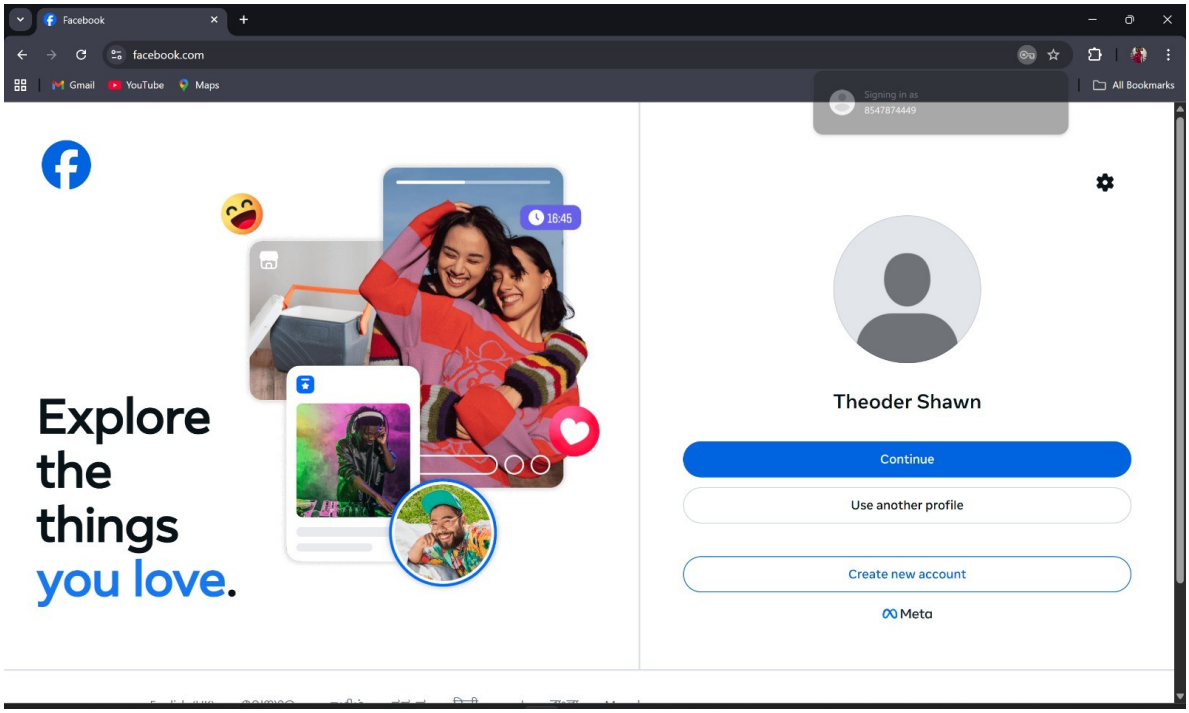
6.2.2 Connected With WIFI Action Required



6.2.3 Web Page For Fingerprint Registration And Deletion

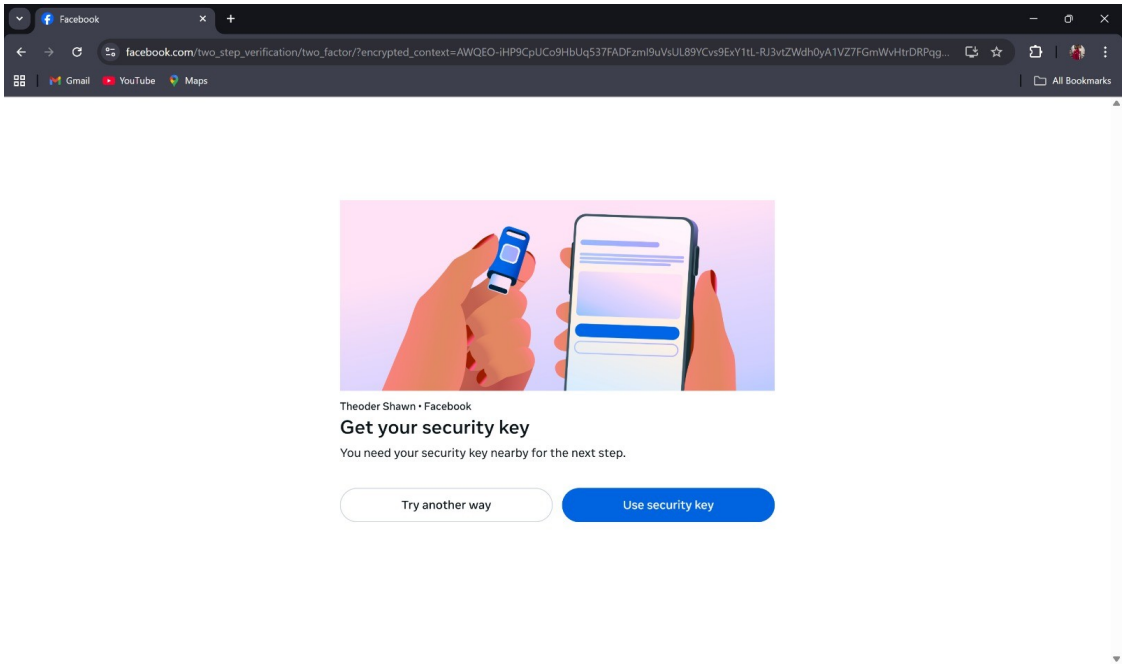
This figure shows the web interface used for fingerprint management through the ESP8266 access point. The *Fingerprint Control* panel allows users to enroll new fingerprints, assign IDs, and delete stored entries. It provides options for single ID deletion or complete reset, ensuring flexibility in managing biometric data. This step demonstrates how fingerprints are securely registered and maintained locally before being used for authentication with the Pico-FIDO device.

6.3 Login Process (2-Factor Authentication)



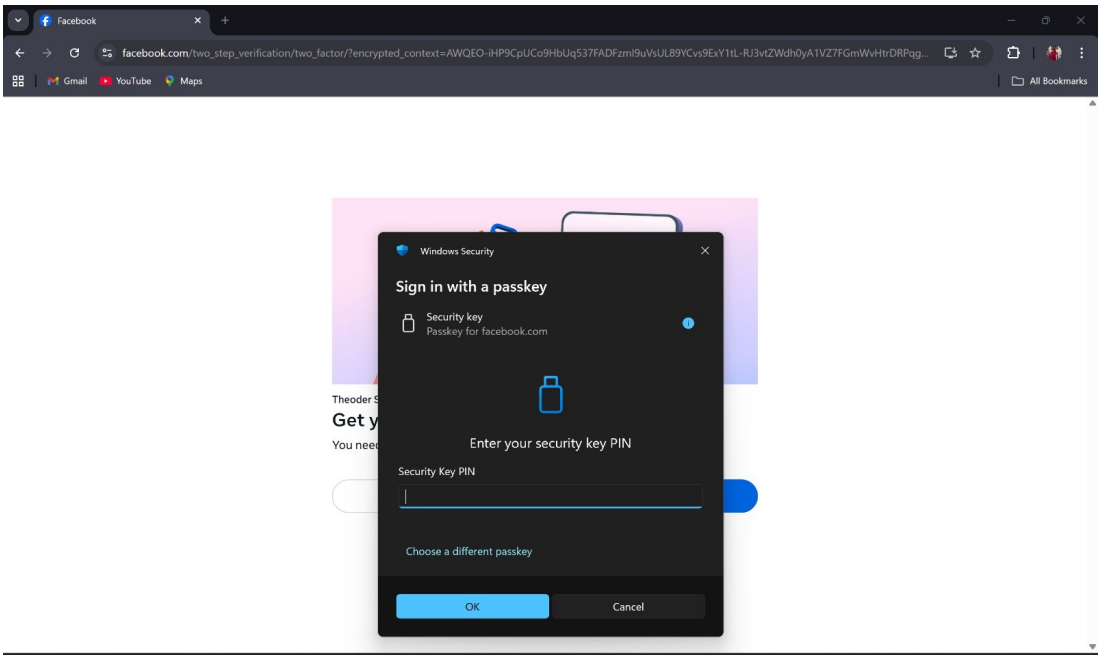
6.3.1 Initial Login with Credentials

This figure shows the initial step of the login process on Facebook, where the user begins authentication using their account credentials. After entering the username and password, the system prompts the user to continue with additional security verification. This stage demonstrates how TouchAuth integrates into the two-factor authentication workflow, requiring the presence of the hardware security key for completing the login.



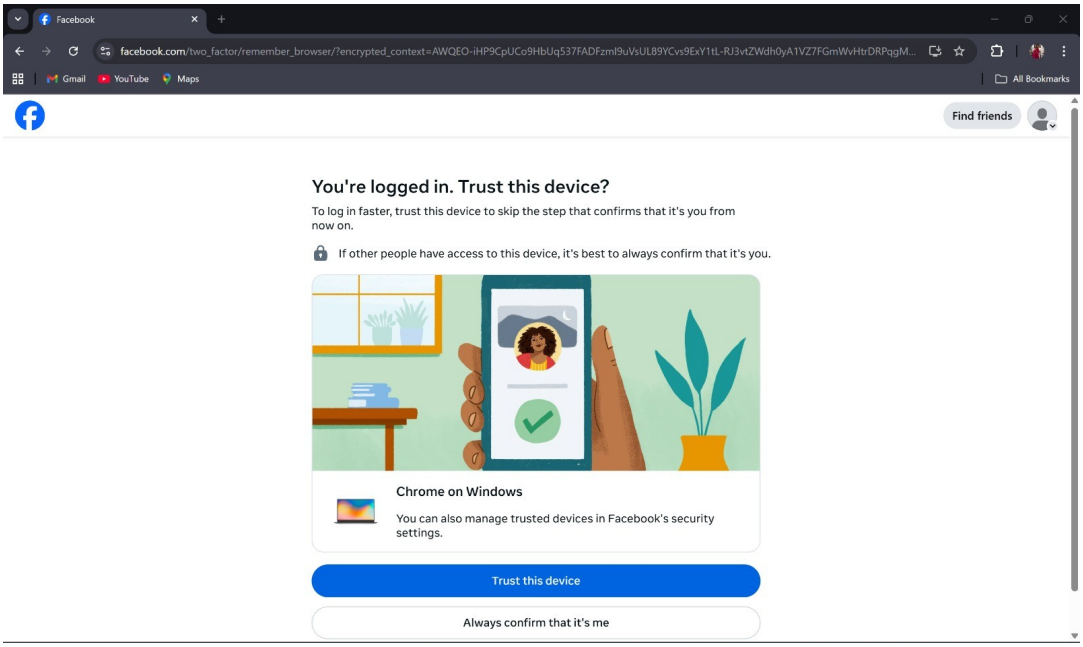
6.3.2 Getting the Security Key

This figure illustrates the second stage of Facebook's two-factor authentication process, where the system explicitly requests the user to insert or use a hardware security key. At this point, the TouchAuth device becomes essential, as it provides the biometric-gated user-presence signal required for FIDO2 authentication. The prompt ensures that login cannot proceed without the physical security key, thereby protecting the account against phishing and unauthorized access. This step demonstrates how TouchAuth integrates seamlessly into existing web authentication flows, reinforcing both usability and strong identity assurance



6.3.3 Security Key PIN Entry

This figure shows the Windows Security prompt during Facebook's two-factor authentication process. After inserting the TouchAuth device, the user is asked to enter the security key PIN to complete login. This step demonstrates how the system combines biometric verification with hardware-based cryptographic authentication, ensuring that only the rightful user can access the account. It highlights the layered security approach of TouchAuth, where both fingerprint validation and PIN entry are required for strong identity assurance.



6.3.4 Login Confirmation

This figure shows the final stage of the authentication process, where Facebook asks whether to trust the current device for faster future logins. For stronger security, the recommended option is to select “Always confirm that it’s me”, ensuring that each login requires verification with the TouchAuth device. This step highlights how the system balances convenience with security, giving users control over how their authentication is enforced.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The completion of this project demonstrates the successful integration of biometric verification, hardware-anchored security, and modern passwordless authentication standards into a unified, high-assurance access control system. By combining a fingerprint-based user-presence mechanism with a dedicated FIDO2-compliant USB authenticator, the design achieves a level of security that far surpasses traditional password-centric models, eliminating the vulnerabilities associated with weak credentials, phishing attacks, and credential reuse. The system's architecture built on the strict separation of biometric processing on the ESP8266 and cryptographic execution on the RP2040 Pico ensures that sensitive fingerprint data never leaves the sensor and that private keys remain securely isolated within the authenticator, creating a layered defense model that is both privacy-preserving and tamper-resistant. Through its detailed workflow, the project showcases how biometric gating, GPIO-based user-presence signaling, and hardware-isolated challenge–response operations can operate in harmony to deliver a seamless authentication experience that is intuitive for users yet uncompromising in its security guarantees. The implementation further highlights the practicality of leveraging open standards such as WebAuthn and CTAP2, enabling the system to integrate effortlessly with modern browsers and operating systems without requiring additional software, drivers, or proprietary middleware.

Beyond its immediate functionality, the project serves as a forward-looking demonstration of how next-generation authentication systems can be engineered to meet the rising demands of enterprise security, cloud platform protection, and secure workstation access. Its modular design allows for scalability, future enhancements, and adaptation to new biometric sensors or cryptographic algorithms, ensuring long-term relevance in an evolving cybersecurity landscape. The system's reliance on physical presence, hardware isolation, and biometric identity creates a robust multi-factor authentication model that is resistant to remote exploitation, credential theft, and unauthorized access attempts.

Moreover, the project illustrates the value of combining embedded systems engineering with modern security standards, showing how microcontrollers, secure protocols, and user-centric design principles can converge to produce a reliable, high-security authentication solution. Ultimately, this work not only validates the feasibility of a biometric-gated FIDO2 hardware authenticator but also establishes a strong foundation for future research, development, and deployment of secure, passwordless authentication technologies in both consumer and enterprise environments.

7.2 Future Scope

The future scope of this project opens the door to a wide range of enhancements that can significantly elevate the system's security, usability, and adaptability in real-world deployments. One of the most impactful advancements would be the integration of a secure element for hardware-backed key storage, enabling the authenticator to store private keys inside a tamper-resistant cryptographic module rather than general microcontroller memory. This would align the system with the security architecture of commercial-grade FIDO2 keys, providing stronger resistance against physical extraction attacks, side-channel analysis, and unauthorized firmware manipulation. Another promising direction is the addition of an onboard display to present real-time authentication status, user prompts, or domain confirmation messages, giving users a more transparent and interactive experience while reducing the risk of phishing by visually verifying the legitimacy of authentication requests. Expanding the system to support multi-user fingerprint profiles would further enhance its versatility, allowing multiple authorized individuals to register their biometric templates and use the same hardware authenticator securely, which is particularly valuable in shared workstations, enterprise environments, or controlled-access facilities.

Beyond user-centric improvements, the system can be extended with a wireless management interface for administrators, enabling secure remote configuration, firmware updates, audit logging, and enrollment management without requiring physical access to the device. This capability would be especially beneficial in large-scale deployments where centralized oversight is essential for maintaining operational integrity and compliance. Strengthening the tamper detection mechanism is another critical area for future development, allowing the device to detect physical intrusion like

attempts, unauthorized opening, voltage manipulation, or abnormal environmental conditions, and respond by wiping sensitive data or locking itself down. Such enhancements would elevate the system from a functional prototype to a hardened security appliance capable of withstanding sophisticated attacks. Collectively, these future improvements demonstrate the scalability and adaptability of the project's architecture, showing that the system can evolve into a fully featured, enterprise-grade authentication platform. By integrating secure hardware modules, improving user interaction, enabling multi-user support, providing administrative control, and enhancing physical security, the project lays a strong foundation for next-generation passwordless authentication solutions that meet the demands of modern cybersecurity landscapes while remaining flexible enough to incorporate emerging technologies and standards.

Looking further ahead, this project also creates opportunities for deeper innovation in both hardware and software layers, enabling the system to evolve into a fully adaptive authentication platform capable of meeting emerging security challenges. Future iterations could incorporate advanced cryptographic algorithms such as post-quantum signatures, ensuring long-term resilience against next-generation computational threats. The architecture could also be extended to support dynamic policy enforcement, where authentication behavior adapts based on contextual factors such as device trust level, network environment, or user risk profile. Integrating secure boot and firmware attestation mechanisms would further strengthen the system by ensuring that only verified, untampered firmware can run on the device, protecting against low-level attacks that target the bootloader or flash memory. Additionally, expanding interoperability with enterprise identity providers and zero-trust frameworks would allow the system to function as a core component in modern organizational security infrastructures, bridging the gap between embedded authentication hardware and cloud-based identity management. By exploring these enhancements, the project can continue to grow beyond its current capabilities, ultimately becoming a comprehensive, future-ready authentication ecosystem that aligns with evolving cybersecurity standards and real-world deployment needs.

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